

Supplementary material

Brain region, head size and helmet fit in on-scalp and cryogenic magnetoencephalography: a simulation study from adults to children

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1 S1. Estimands and conventions

2 The primary metric is the known-topography detectability

$$d = \sqrt{s^T C^{-1} s}, \quad (\text{S1})$$

3 where s is the field pattern of a source across an array's channels and C is that
4 array's noise covariance; d is the output signal-to-noise ratio (SNR) of the optimal
5 matched filter for a source of known topography. We computed it with the true
6 (oracle) covariance and rank-aware whitening for every target and every array, and
7 compared arrays target by target. The adult comparison reports the median over
8 targets of the paired ratio $d_{\text{OPM}}/d_{\text{Neuromag}}$, not the ratio of the two systems' median
9 detectabilities.

10 The same quantity in decibels is $D = 20 \log_{10}(d_{\text{OPM}}/d_{\text{Neuromag}})$, positive when
11 the optically pumped magnetometer (OPM) array is ahead. D does not depend on
12 the dipole moment. Unless stated otherwise, the comparator is all 306 channels
13 of the Neuromag system, a magnetoencephalography (MEG) system of supercon-
14 ducting quantum interference devices (SQUIDS).

15 The adult analyses report the unweighted median over all 7,661 targets, the
16 558 on the medial wall included. The smaller heads use an area-weighted median
17 over cortical targets with the medial wall excluded, because their cortices differ in
18 size and their targets in density. Both medians summarize the same metric. At the
19 adult's measured head position, the dense array's ratio is 1.144 unweighted over
20 all targets, 1.148 without the medial wall and 1.158 area-weighted without it. Its
21 D is +1.17 dB under the adult convention and +1.27 dB under the pediatric one.
22 A 1 % change of the ratio is 0.086 dB, and 1 dB is a ratio of 1.122. Table S1 gives
23 the adult's primary comparison in both forms, and Table S2 lists every estimand
24 of the study, its definition, its interval and what that interval covers.

25 Every Δ is taken against the adult placed by the same rule, because the adult's
26 own D depends on the placement. It is +1.27 dB at its measured head position,
27 +1.00 dB at top contact and +1.01 dB laterally centered and raised to contact. It is
28 +0.87 dB at true contact, +1.27 and +1.28 dB in its own helmets scaled with the
29 head (centered and laterally centered; their scale factor is 1.000), and +1.28 dB
30 in the helmet fitted at its own gap. Where a difference of two pooled medians is

Table S1. The adult’s primary comparison as a ratio and in decibels. Median over all 7,661 targets of the paired ratio OPM/Neuromag (306 channels), with 95 % parcel-bootstrap intervals in brackets; OPM white noise 15 fT/√Hz; the adult at its measured head position.

Array and form	Sensor noise only	Sensor plus brain noise (primary)	Room field projected out
Dense, ratio	0.742 [0.717, 0.768]	1.144 [1.121, 1.168]	1.115 [1.081, 1.149]
Dense, D (dB)	-2.59 [-2.89, -2.30]	+1.17 [+0.99, +1.35]	+0.95 [+0.68, +1.21]
Site-matched, ratio	0.531 [0.512, 0.551]	1.009 [0.994, 1.027]	0.948 [0.899, 0.977]
Site-matched, D (dB)	-5.50 [-5.82, -5.18]	+0.08 [-0.05, +0.23]	-0.46 [-0.92, -0.20]

Dense, the dense OPM array; site-matched, the site-matched OPM array; $D = 20 \log_{10}$ of the ratio.

Table S2. Estimands of the study in full. d , known-topography detectability with the oracle covariance; $D = 20 \log_{10}(d_{\text{OPM}}/d_{\text{Neuromag}})$; Neuromag means all 306 channels unless stated. Every parcel bootstrap is a percentile interval over the parcels of one anatomy; it describes spatial variation within that anatomy and excludes placement, construction and between-subject variability.

Estimand	Definition	Interval and what it covers	Used in (section)
Adult ratio	Median over all 7,661 targets (medial wall included) of the paired ratio $d_{\text{OPM}}/d_{\text{Neuromag}}$, or D ; conditions: sensor noise only, sensor plus brain noise (primary), with the room field, projected	95 % bootstrap over the adult’s 70 labels (atlas parcels and the two medial-wall labels)	3.1–3.3; S2, S3
Scenario ratio	The adult ratio with sensor, brain and room noise (Neuromag’s 305 good channels) under one of five stated assumptions about how each system’s real noise departs from the model, informed by Neuromag’s measured covariance (scenarios A–E)	Parcel bootstrap over the same labels	3.2; S3
Break-even noise	The OPM white-noise level at which the adult ratio equals 1, found on a grid of 33 levels over 4–60 fT/√Hz	Parcel bootstrap of the crossing	3.1, 3.2; S3
Depth-bin ratio	Median of the paired ratio over the targets of one 5-mm depth bin	Bootstrap over the parcel labels present in the bin	3.3
Regional ratio	Median of the paired ratio over a lobe’s or a parcel’s targets	Lobes: bootstrap over the lobe’s parcels; single parcels: the interquartile range over targets, a spread rather than a confidence interval	3.3
D of a head	Area-weighted median of D over a head’s cortical targets, medial wall excluded, in a named helmet and placement	Parcel bootstrap within the head	3.4, 3.5; S5, S6
Δ	$D_{\text{head}} - D_{\text{adult}}$, the adult placed by the same helmet rule: vertex-wise for the scaled adults (the same cortical vertex); for the templates and children, which have no vertex correspondence, the adult-area-weighted median of parcel-level differences	Parcel bootstrap	3.4, 3.5; S5

Table S2. Estimands of the study in full (continued).

Estimand	Definition	Interval and what it covers	Used in (section)
Δ within a depth stratum	The difference of the two heads' area-weighted medians within one stratum of depth below each head's own scalp (edges at 0, 10, 15, 20, 25, 30, 40, 50, 60, 90 mm), for every head including the scaled adults; strata with fewer than 10 targets in either head are not summarized	Parcels resampled in each head independently	S5
Depth-reweighted difference	The difference of pooled medians after weighting each head's targets to the adult's depth distribution; not the same estimand as Δ	None (point estimate)	S5
Within-head contrast	The area-weighted median over one head's targets of the target-wise difference $D(\text{fixed helmet at top contact}) - D(\text{fitted helmet})$, not the difference of the two medians; the OPM array is the same in both, so it is a property of Neuromag alone	Parcel bootstrap	3.5; S5
Interaction	The within-head contrast of the smaller head minus that of the adult; at every vertex of a scaled adult it equals $\Delta(\text{fixed}) - \Delta(\text{fitted})$, but the area-weighted medians are not additive	Parcel bootstrap	3.5, 4.2, 5; S5
Δ at equal OPM site count	Δ with the adult's dense array subsampled by farthest-point sampling to the smaller head's dense site count, the adult in the same helmet rule	Parcel bootstrap	3.5; S5
Placement band	Δ at each of the 12 source-blind placements of the fixed helmet (each head against the adult at the same placement), as a difference of area-weighted medians; the crossed band pairs any feasible placement of the head with any of the adult	The range over placements, no interval	3.4; S5

Table S2. Estimands of the study in full (continued).

Estimand	Definition	Interval and what it covers	Used in (section)
Peak-channel SNR	Each system's largest ratio of a channel's noise-free signal to that channel's own in-band noise standard deviation; OPM over Neuromag (best of all channels, or of the magnetometers)	Parcel bootstrap (adult); medians only (smaller heads)	3.1, 4.3; S2, S3, S6
Mean-power SNR	$10 \log_{10}$ of the channel mean of the squared SNRs (Goldenholz et al., 2009), compared as an amplitude ratio	Parcel bootstrap	3.1
Plug-in detectability	The output SNR under the true covariance of the matched filter built from a Ledoit–Wolf estimate from 780 or 4,680 independent samples (twice the time-bandwidth product of 10 and 60 s in this band), drawn for all arrays from one common noise realization	Parcel bootstrap	3.1; S3
S50	The strength at which a depth band's focal detection rate reaches 50 %, interpolated linearly in log strength; the band's locations (18 per band in the exploratory run, 36 in the confirmatory run) and 3 morphologies are pooled	Bootstrap over locations (1,000 resamples); open at one end when resamples do not reach half detection within the tested strengths	3.6; S7
Paired S50 ratio	$S50_{\text{Neuromag}}/S50_{\text{OPM}}$, a ratio of the two pooled S50 estimates, above 1 when the OPM needs less strength; each bootstrap resample draws locations with replacement and recomputes both S50 values on the same locations, so the ratio is paired by location and is not a median of location-specific ratios	Location bootstrap, as S50	3.6; S7
Detection-count difference	Per location, the number of events detected by one array minus the other, tested with the two-sided sign-flip test over locations; every p-value reported is exact over all sign patterns	None; a p-value	3.6; S7
Localization error	Distance from the true source (a patch's center) to the fitted dipole or to the dSPM peak, read at the true peak sample through the perturbed transform	Paired differences: bootstrap over events; Wilcoxon signed-rank p-value	3.6; S7
Joint success	Detection by the practical detector with a localization error of at most 10 mm; paired differences tested with an exact McNemar test	None; a p-value	S7

Sections numbered 3.1–5 are those of the main text; sections S2–S7 are those of this supplementary material. dSPM, dynamic statistical parametric mapping (Dale et al., 2000); SNR, signal-to-noise ratio.

31 used instead of Δ , the text says so. Δ is reported descriptively, without a test or a
32 margin.

33 Peak-channel SNR lets magnetometers and gradiometers compete on equal
34 terms, because each channel is divided by its own noise. It is the closest analogue
35 of the sensor-level SNRs of the clinical studies, with one difference: a clinical base-
36 line standard deviation holds physiological noise as well as sensor and room noise,
37 whereas the model's holds only what the noise model contains. Mean-power SNR
38 is the channel-averaged SNR of Goldenholz et al. (2009), also used by Matsubara
39 et al. (2026).

40 The plug-in filter is built from a Ledoit–Wolf covariance estimate (Ledoit and
41 Wolf, 2004) and evaluated under the true covariance, so it cannot exceed the oracle
42 value. The estimate used 780 or 4,680 independent samples, the time-bandwidth
43 equivalents of 10 and 60 s of data in this band, drawn for all arrays from one com-
44 mon noise realization. The estimate from 4,680 samples is the declared 60-s sce-
45 nario; its samples are independent, which does not establish the effective sample
46 size of a measured colored-noise baseline. The estimate from 780 samples shows
47 how a short baseline penalizes the array with more channels.

48 Neighboring cortical targets are correlated, so no interval resamples targets as
49 if they were independent. The one exception is the adult's frequency-band compar-
50 ison, which resamples targets and therefore has narrower intervals than the primary
51 comparison. All intervals are percentile intervals. In the spike study the location
52 is the statistical unit: S50 ratios carry location-bootstrap intervals, and detection
53 differences are tested with the sign-flip test on per-location differences in detection
54 counts, reported exactly over all sign patterns (Section S7). Table S16 gives the
55 resample count of every interval family.

56 **S2. The spherical benchmark**

57 We reimplemented the analytical model of Jas et al. (2026) with their defini-
58 tions. A tangential current dipole of 30 nAm sits in a spherical conductor (head
59 radius 95 mm, brain radius 80 mm), and point magnetometers measure the radial
60 field on the scalp (OPM) and 18 mm above it (SQUID). The single-channel SNR
61 is the peak field over the noise standard deviation σ , and $\sigma_{\text{OPM}} = \eta \sigma_{\text{SQUID}}$. The
62 equal-SNR depth is the exact root of the preprint's Eq. 3. We checked the peak
63 field against the full Sarvas field maximized over the sensor sphere (largest rel-
64 ative difference 4.7×10^{-15}) and against the sphere model of MNE-Python (5.4
65 $\times 10^{-8}$; Gramfort et al., 2013).

66 The reimplementation reproduced the preprint. At $\eta = 3$ the exact equal-SNR
67 depth is 27.665 mm. The preprint prints 28 mm in its text and caption, and its
68 figure draws 27.53 mm; the difference reflects the preprint's rounding and grid,
69 not a difference of model. A crossing exists only for η between 1.683 and 5.309

(printed as 1.7 and 5.3). Only η enters the equal-SNR depth: scaling both noises together leaves it at 27.665 mm.

On the real head, the peak field at the best OPM over that at the best Neuromag magnetometer falls with depth as the sphere predicts, once the sphere is given the real median standoffs: 7.0 mm for the OPMs and 29.8 mm for the magnetometers. For the dense array this ratio fell from 5.41 in the 10–15 mm bin to 2.28 at 55–60 mm, against 5.26 and 2.09 for the sphere.

The bridge between the two models uses the sphere’s own metric: peak-channel SNR against the magnetometers, with sensor noise only. With it, the realistic equal-SNR depth at $\eta = 3$ was 29.6 mm for the site-matched array and 31.8 mm for the dense array, against 29.0 mm for the sphere with the real standoffs. For η from 2.5 to 4.5, the real arrays’ depths lay within -0.8 to $+5.4$ mm of the sphere’s (Fig. S1B). Outside that window they differed.

The OPMs were ahead at every depth for $\eta \leq 2.25$, where the sphere still predicts a deep crossing (48.8 mm at $\eta = 2.25$). The reason is that the real peak-field ratio levels off at depth (2.35 at 60–65 mm), while the sphere’s keeps falling (2.02). Neuromag’s magnetometers were ahead at every depth for $\eta \geq 5.0$ (site-matched) or 5.5 (dense). The sphere is therefore right about the geometry of magnetometers near its reference η .

Against Neuromag’s 204 planar gradiometers, the ratio with sensor noise only barely changed with depth, so the sphere describes only the magnetometer part of the comparison. What it leaves open is η itself, which the explicit noise model of Section S3 replaces.

For head size, we placed the newborn, 1-year, 8-year and adult spheres of the preprint in a SQUID shell that follows each head at 18 mm, as in the preprint. As an additional contrast, we placed all of them in the adult shell of radius 113 mm. With the size-following shell, the equal-SNR depth at $\eta = 3$ hardly moved from adult to newborn (27.7–30.8 mm). The normalized depth, however, rose from 15.8 % to 49.6 %, because the brain and the extracerebral layer (7–15 mm in the preprint’s heads) shrink.

With one adult shell for every head, size and fit are confounded. The OPM was then ahead at every depth of the newborn and 1-year brains up to $\eta = 8.7$ and 4.2, respectively. The sphere’s prediction for children is thus conditional on a shell that follows the head at a constant gap.

S3. The noise model: components, calibration, validation and sensitivity analyses

S3.1. Components and calibration

Every array saw identical noise sources in one analysis band of 1–40 Hz (zero-phase Butterworth filter of order 4; equivalent noise bandwidth 35.1 Hz). The

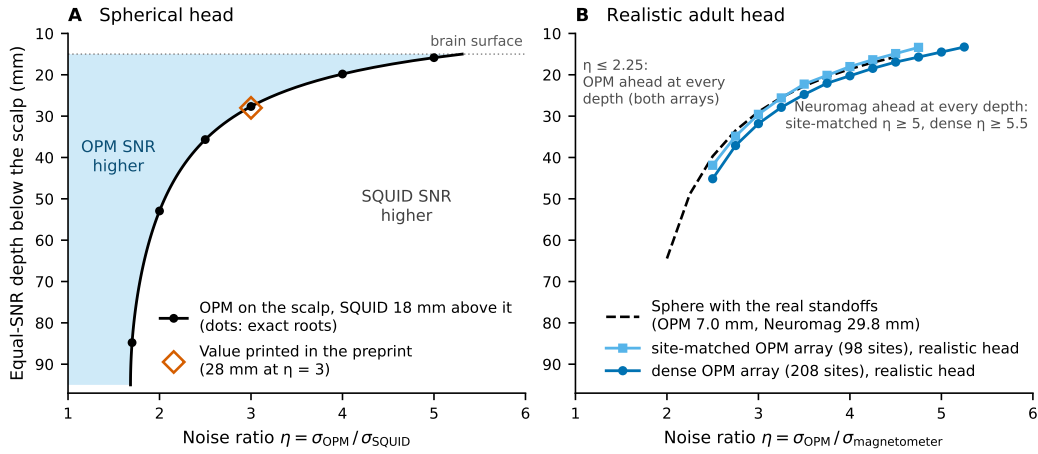


Fig. S1. The spherical benchmark and its transfer to a real head. (A) The spherical head of Jas et al. (2026), reimplemented: the equal-SNR depth below the scalp against the noise ratio $\eta = \sigma_{\text{OPM}} / \sigma_{\text{SQUID}}$ for a point magnetometer on the scalp (OPM) and one 18 mm above it (SQUID). Dots, exact roots of the preprint’s Eq. 3 at selected noise ratios; diamond, the value the preprint prints, 28 mm at $\eta = 3$; dotted line, the brain surface, 15 mm below the scalp (head radius 95 mm minus brain radius 80 mm); shaded, the OPM’s single-channel SNR is higher. Panel A is adapted from Jas et al. (2026), Fig. 4E (CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>), redrawn from our reimplement of their model. (B) The same on the realistic adult head with the sphere’s own metric, peak-channel SNR against the Neuromag magnetometers with sensor noise only, for the site-matched array (98 sites, squares) and the dense array (208 sites, circles), drawn at the noise ratios at which each array has a crossing. Dashed, the sphere at the real median standoffs (7.0 mm for the OPMs, 29.8 mm for the magnetometers). For $\eta \leq 2.25$ both arrays are ahead at every depth; Neuromag’s magnetometers are ahead at every depth for $\eta \geq 5.0$ (site-matched) and $\eta \geq 5.5$ (dense). This bridge is a single-channel quantity against magnetometers only, not the primary endpoint.

noise had three components: sensor noise, the cortical background and the room field.

Sensor noise was white: 3.5 fT/ $\sqrt{\text{Hz}}$ for the Neuromag magnetometers and 3.6 fT/(cm $\sqrt{\text{Hz}}$) for the gradiometers, the typical values of the TRIUX datasheet (Elekta Neuromag, n.d.), and 15 fT/ $\sqrt{\text{Hz}}$ for the OPMs in the primary analysis, swept over 7–30 fT/ $\sqrt{\text{Hz}}$. The OPM value is a declared parameter, not a measured device value. It is the middle of the sweep and lies within the range of sensor noise quoted by the clinical OPM studies, although those studies state no band for these figures and the device specifications behind them begin at 1 Hz or above (Osborne et al., 2018). The top of the sweep is close to the empty-room floor read from the recording of Jas et al. (2026), a reading of its Fig. 9 that assumes its unstated dB reference is 1 fT²/Hz. We report the comparisons across the sweep.

The cortical background consisted of 1,755 independent dipoles normal to the cortex on a 7-mm grid, weighted by cortical area. Like the targets, it omitted the cortex within 4 mm of the inner skull (8.7 % of the vertices), which holds the sources nearest to every sensor; Section S3.6 adds that cortex back. One scale was fitted so that the median good gradiometer’s in-band variance equaled the measured brain noise of the sample recording (main text, Section 2.4). This brain noise is the in-band variance of the pre-stimulus windows of its 320 stimuli minus that of the empty-room recording, the channel marked bad excluded. The windows begin soon after the previous stimulus, so they hold task-related rather than only resting activity. Spatially correlated variants used correlation lengths of 5 and 10 mm. The room field was an external field of 8 terms (uniform plus linear gradient), fitted to the empty-room recording and passed through each array’s coil integration.

Each array’s oracle covariance was computed in closed form, without sampling:

$$C = \Sigma_s + s^2 L_{\text{bg}} A L_{\text{bg}}^T + E \Sigma_e E^T, \quad (\text{S2})$$

where Σ_s is diagonal, each entry being a channel’s squared white-noise spectral density times the filter’s equivalent noise bandwidth (35.1 Hz); L_{bg} holds the array’s lead fields of the 1,755 background dipoles; $A = \text{diag}(a_i)$, a_i being the area of source i ’s Voronoi cell on the cortex; E holds the array’s responses to the external-field terms; and Σ_e is their coefficient covariance. The background dipoles are independent and cortex-normal, a greedy Poisson-disk subset of the usable white-surface vertices with 7-mm minimum spacing (seed 2026). Their Voronoi cells partition the 1,878 cm² of usable cortex (107 mm² on average).

Source i ’s moment variance is $s^2 a_i$, with $s^2 = 0.0354$ nAm² per mm². Its standard deviation is 0.188 nAm $\times \sqrt{(a_i/\text{mm}^2)}$, or 1.95 nAm root-mean-square (RMS) over the sources. The scale s^2 was fitted once, so that the good gradiometers’ median modeled variance equals their median measured brain noise (37.1 fT/cm

148 RMS), and was kept per unit area on every head. The matrix E is computed through
 149 each array's own coil integration points, for 8 external-field terms (3 uniform and
 150 5 linear-gradient, about (0, 0, 40) mm in the head frame); Σ_e was fitted to the
 151 band-filtered empty-room recording. Plug-in covariances are Ledoit–Wolf esti-
 152 mates from 780 or 4,680 independent samples, twice the time-bandwidth product
 153 of 10 and 60 s.

154 The room field is removed by the projection

$$P = I - E (E^T W E)^{-1} E^T W, \quad (\text{S3})$$

155 where $W = \Sigma_s^{-1}$. It is applied to topographies and covariance alike (P_s , PCP^T),
 156 by the same rule for every array. For Neuromag it acts on all 306 channels jointly,
 157 so its magnetometer and gradiometer comparisons after projection use subsets of
 158 jointly projected data. The retained ranks were 298 of 306 for Neuromag (99 of 102
 159 magnetometers, 204 of 204 gradiometers), 200 of 208 for the dense array and 90
 160 of 98 for the site-matched array. Whitening drops eigenvalues of the correlation-
 161 normalized covariance below 10^{-10} of the largest. Relative to the room-field con-
 162 dition without projection, the projection cost Neuromag 0.48 % of its detectability,
 163 the dense array 2.5 % and the site-matched array 6.0 %.

164 Each OPM's sensitive axis was the normal of the boundary-element-method
 165 (BEM) head surface, placed on the MRI scalp with normals from its own trian-
 166 gles, averaged within 15 mm of the site. It deviated from a plane fitted to the
 167 dense MRI scalp by a median 1.2° (95th percentile 4.9° site-matched, 7.4° dense).
 168 An alternative construction took the adult's stored outer-skin normals, which lie
 169 within 5° of the radial direction. With those axes at the same sites, the projected
 170 dense ratio was 1.073 [1.017, 1.115] instead of 1.115, and 0.82 instead of 0.98 at
 171 45–70 mm. Nearly radial axes make room-field patterns resemble those of deep
 172 sources, so the projection removes more of their signal. With sensor plus brain
 173 noise and no projection, the two constructions gave 1.130 against 1.144.

174 The Neuromag model combines the sample recording's VectorView geometry
 175 (102 magnetometers, 204 planar gradiometers) and the T3 coil definitions of MNE-
 176 Python (types 3014 and 3024; accurate integration over 8 points per gradiometer
 177 and 16 per magnetometer). Its white noise is the manufacturer's typical value
 178 for the TRIUX system, 3.5 fT/ $\sqrt{\text{Hz}}$ and 3.6 fT/(cm $\sqrt{\text{Hz}}$) (20.7 fT and 21.3 fT/cm
 179 in the band), with the 18-mm coil-to-helmet spacing. These three values are the
 180 typical values of the TRIUX datasheet (Elekta Neuromag, n.d.): they describe a
 181 representative Neuromag system, not one installation. Noise from the measured
 182 empty-room spectrum (25.9 fT, 20.0 fT/cm) is a sensitivity analysis. No signal-
 183 space separation is applied.

184 In the band sensitivity analysis (1–10, 8–30 and 30–80 Hz), the brain scale and
 185 the room field were refitted in each band, with and without a first-order 100-Hz

186 OPM response. The dense ratio with sensor plus brain noise was 1.15 at 1–10 Hz,
187 1.14 at 8–30 Hz and 1.11 at 30–80 Hz (1.07 with the OPM response), against 1.14
188 in the analysis band. The site-matched ratio was 1.00, 1.01 and 0.99. Sensor noise
189 was 1.3 % of a dense-array channel’s variance at 1–10 Hz and 23 % at 30–80 Hz,
190 and 9.3 % and 74 % of a Neuromag channel’s. We chose a 1–40 Hz band because
191 the matched filter’s gain rests on the lower frequencies, where the brain noise sits.

192 In a 3–70 Hz band, typical of clinical spike review, the dense ratio with sensor
193 plus brain noise was 1.137 (1.130 with the OPM response) and the site-matched
194 ratio 1.011, against 1.144 and 1.009 in the analysis band. Their target-bootstrap
195 intervals, [1.133, 1.141] and [1.009, 1.013], are narrower than the parcel bootstrap
196 of the primary comparison (Section S1). With the room field projected out, the two
197 ratios were 1.108 and 0.945. In that band, sensor noise was 5.6 % of a dense-array
198 channel’s variance and 34 % of a Neuromag channel’s.

199 *S3.2. What is fitted, calibrated, checked and predicted*

200 Four kinds of agreement must be kept apart (Fig. S2). For the magnetometers,
201 the room field was fitted to the empty-room recording. With the manufacturer’s
202 sensor noise it reproduced their empty-room level (114.9 against 115.2 fT mea-
203 sured) and explained 94 % of their empty-room variance. For the gradiometers
204 the room field explained only 1.6 % of that variance. Their empty-room agree-
205 ment (21.4 against 20.2 fT/cm measured) is therefore an independent check of the
206 model’s sensor-noise term, the datasheet’s typical value, against this one record-
207 ing, whose level it slightly exceeds. The gradiometer brain noise is calibrated:
208 37.1 fT/cm, equal to the measurement by construction.

209 The magnetometer brain noise is the one independent prediction of the back-
210 ground at the level of a median, and it fell short: 192 fT modeled against 262 fT
211 measured (0.73 times). The measured value is the excess of the task baseline over
212 the empty room, which also holds non-cortical physiological fields that the model
213 omits by design. The shortfall is thus not purely an error of the cortical back-
214 ground; Section S3.4 separates the heart’s part. Nothing about the OPM noise
215 was measured: its sensor noise is declared, and its brain and room noise are the
216 Neuromag-calibrated model seen through the OPM arrays’ own lead fields.

217 Two published simulation studies check shape and level. The depth-orientation
218 maps of Hunold et al. (2016), adapted to this anatomy, correlated with the pub-
219 lished dipole maps at $r = 0.96$ – 0.98 but reached only 0.74–0.79 times their level
220 (patch maps: $r = 0.93$ – 0.94 , 0.79–0.85 times), even with a background scale fitted
221 to the paper’s example traces. The calibration rule of Goldenholz et al. (2009)
222 gave a noise-source amplitude within their published range once normalized to
223 their number of sources (1.77 against 1.6–1.9 nAm; 2.68 nAm on our grid). Each
224 adaptation keeps its paper’s design rather than this study’s model. For Hunold

et al. (2016) these are 600-nAm spike-wave sources over a background of randomly placed cortical dipoles, conductivities of 0.33, 0.0042, 0.33 S/m and their peak-to-peak SNR. For Goldenholz et al. (2009) they are 10-nAm dipoles, channel-averaged power SNR and a cortical background calibrated on recorded noise.

Both adaptations already show the dependence on metric that the main text examines. With brain noise only, the best single channel of a site-matched OPM array in our adaptation of the dipole maps of Hunold et al. (2016) reached only 0.70–0.84 times the SNR of the planar gradiometers in every depth row. It reached 0.89–1.37 times that of the magnetometers, falling with depth. With the channel-averaged SNR of Goldenholz et al. (2009), by contrast, the OPM was within about 0.1 dB of both (point estimates: -0.08 dB against the gradiometers and $+0.09$ dB against the magnetometers).

In this model brain noise dominates both kinds of magnetometer channel. It was 92 % of a dense-array OPM channel’s in-band variance (room field 4.9 %, sensor noise 3.0 %) and 74 % of a Neuromag magnetometer’s (room field 25 %, sensor noise 0.9 %), whereas sensor noise was 25 % of a gradiometer’s (Fig. S2A and C). In RMS terms, a dense-array channel carried 494 fT of brain noise, 114 fT of room field and 89 fT of sensor noise in the band, against 192, 113 and 21 fT at the magnetometers.

The implied single-channel noise ratio OPM/magnetometer, the analogue of η , was about 2.6 in the 1–40 Hz band with sensor plus brain noise, the primary condition (2.3 with the room field added). In the recording of Jas et al. (2026) it is about 4.6, over a much wider band (4–330 Hz). Approximate readings of the preprint’s spectra, not results of this study, locate the difference. Where brain activity dominates, the OPM/SQUID ratio of the spectral peaks is about 2.0, somewhat below the model’s 2.6. The measured OPM, however, has an empty-room floor of about 28–35 fT/ $\sqrt{\text{Hz}}$ and a distance-independent low-frequency component of about 200–224 fT/ $\sqrt{\text{Hz}}$ at its peak. On this reading, the model lacks the OPM’s own floor and a non-brain low-frequency component, which would explain why the measured OPM noise barely fell when the sensors were lifted. The ratio of the spectral peaks and the floor are uncertain readings of the preprint’s Fig. 9, and the preprint (version 1) has not been peer reviewed.

S3.3. Why the ratio falls with depth, and the other adult conditions

Against Neuromag’s 306 channels, the fall of the dense ratio with depth was already present with sensor noise alone, from 1.00 to 0.64. Neuromag’s detectability with sensor noise only rests on its magnetometers, and the peak-field ratio OPM/magnetometer fell from 5.41 in the shallowest bin to 2.28 at 55–60 mm. The detectability ratio fell with it, though less steeply, since the whole-array measure draws on the field at many channels.

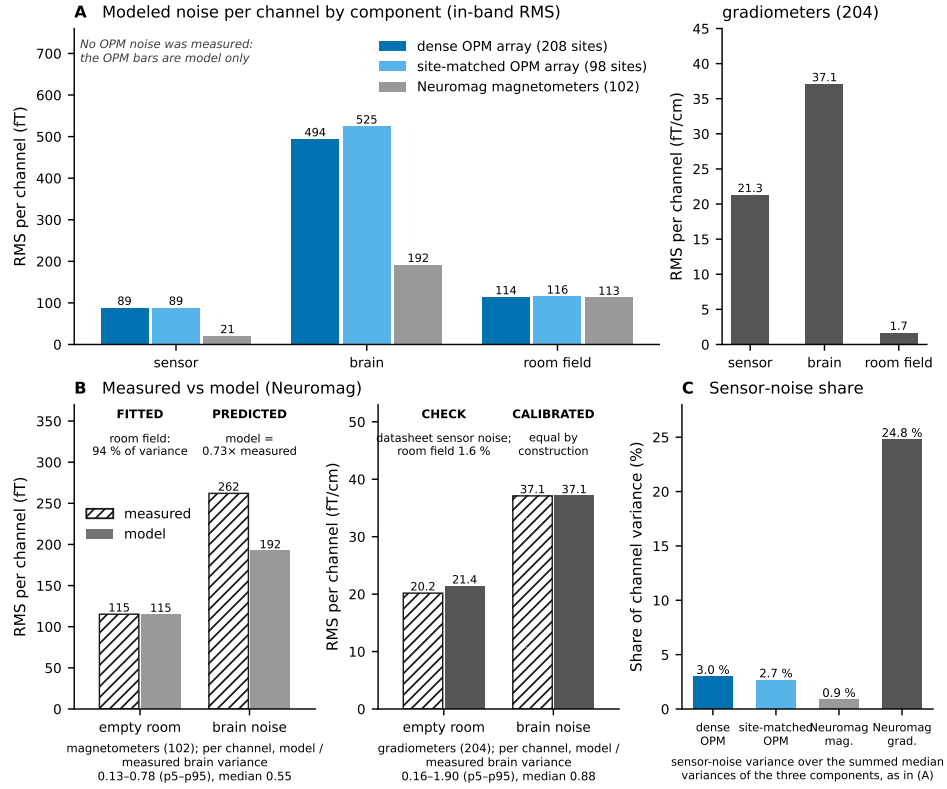


Fig. S2. The noise budget and what is fitted, checked, calibrated and predicted. (A) Modeled in-band (1–40 Hz) RMS noise per channel by component, the square root of each component’s median channel variance (RMS values do not add): sensor noise (15 fT/√Hz for the OPMs, the TRIUX datasheet’s typical values for Neuromag), cortical background and room field, for the dense OPM array (208 sites), the site-matched OPM array (98 sites) and Neuromag’s 102 magnetometers; the 204 gradiometers on their own fT/cm axis. No OPM noise was measured: the OPM bars are model only. (B) The measured quantities of the Neuromag recording against the model. Magnetometers: the empty-room level, a fit (the room field explains 94 % of its variance), and the brain noise, the task baseline minus the empty room, a prediction (0.73 times the measurement). Gradiometers: the empty-room level, a check of the datasheet’s sensor noise (the room field explains 1.6 %), and the brain noise, calibrated and therefore equal by construction. The labels under the panels give the 5th to 95th percentiles over channels, and the median, of the modeled-to-measured brain-noise variance per channel, for the 102 magnetometers and 204 gradiometers of the adult comparison; the covariance check of Section S3.4 leaves the bad channel out, so its gradiometer range differs slightly. (C) The sensor-noise share of the in-band variance with brain noise and room field, for the two OPM arrays and for Neuromag’s magnetometers and gradiometers. RMS, root-mean-square.

264 Against the 204 gradiometers alone, the ratio with sensor noise only barely
265 changed with depth (3.56 at 10–15 mm, 3.08 at 20–25 mm and 3.44 at 60–65 mm).
266 With brain noise, however, it fell from 1.88 to 1.17 at 35–40 mm. Here the depth
267 dependence comes from brain noise, which costs every array most for deep sources.

268 Against all 306 channels, adding brain noise multiplied the dense array’s ratio
269 by a nearly constant factor in every depth bin (1.51–1.68). For the site-matched ar-
270 ray the factor rose with depth (1.69–2.09) and partly offset the geometric fall. The
271 size of the factor follows the per-channel noise budget rather than better whiten-
272 ing by the denser array. Brain noise multiplied the in-band noise of a Neuromag
273 magnetometer about 9.3-fold over its sensor floor but that of an OPM only about
274 5.7-fold (a ratio of 1.65), and the sparser site-matched array had the larger factor.

275 With sensor plus brain noise, the dense array was behind Neuromag at 4 of the
276 7,661 targets. They lie on the left medial orbitofrontal (3) and left medial wall (1),
277 26.5–35.1 mm below the scalp, with ratios of 0.956–0.997. Pooled over both hemi-
278 spheres, 12 atlas parcels had a smaller dense lead than the parahippocampal cor-
279 tex (1.12): posterior cingulate (1.03), precuneus (1.05), caudal anterior cingulate
280 (1.05), transverse temporal (1.06), lingual (1.06), pericalcarine (1.06), paracentral
281 (1.07), rostral anterior cingulate (1.07), isthmus cingulate (1.07), insula (1.09),
282 medial orbitofrontal (1.12) and banks of the superior temporal sulcus (1.12).

283 A covariance estimated from 780 samples cost Neuromag more than the OPMs
284 (15 % of the oracle detectability, against 9.1 % for the dense array and 5.1 % for
285 the site-matched array). It raised the dense ratio to 1.23 and the site-matched ratio
286 to 1.12. With 4,680 samples, the declared 60-s scenario, the penalty was small for
287 every array (main text, Fig. 2). The site-matched array’s interval lay above 1 at an
288 OPM white noise of 10 fT/√Hz (1.049 [1.032, 1.068]) and below 1 at 20 fT/√Hz
289 (0.983 [0.967, 0.997]).

290 As a triaxial control, three axes with the same noise on every axis at the 98 site-
291 matched sites (294 channels) raised the ratio from 1.01 to 1.12 (1.17 projected).
292 This is close to the dense array’s 1.14 from 208 single-axis sites. If the tangential
293 axes are twice as noisy, the triaxial ratio is 1.05.

294 *S3.4. The model against the measured Neuromag covariance*

295 The measured brain noise is the covariance of the sample recording’s 320 pre-
296 stimulus windows in the 1–40 Hz band minus that of the empty-room recording.
297 As in the adult comparison, the bad channel (MEG 2443) was left out and no
298 signal-space projection was applied; 305 channels remain (102 magnetometers,
299 203 gradiometers). Two quantities of the model are fitted to this dataset: the one
300 scalar of the cortical background (the median good-gradiometer brain variance)
301 and the room field’s 8×8 coefficient covariance (the empty room). Everything
302 else was compared as an independent prediction. This covers the magnetometers’
303 brain-noise amplitude and channel pattern, the gradiometer level at held-out sites,

304 the spatial correlations against inter-sensor distance, the eigenvalue spectra and
305 dominant subspaces, the sub-band spectra, and Neuromag's detectability with the
306 measured covariance.

307 Where a sample covariance is compared with an exact one, the comparison
308 is calibrated on surrogates. An exact model "measured" the same way (phase-
309 randomized records recolored to the model, with the same number of samples)
310 gives the range that an exact model would show. A value outside that range is
311 therefore a real departure of the measured noise from the model.

312 The model predicted the magnetometers' median brain-noise amplitude at 0.73
313 of the measured; an exact model would give 0.98–1.03. For the gradiometers, the
314 scale fitted on half the sites predicted the other half's median amplitude to 1.00
315 [0.93, 1.09] over 1,000 random halves (Fig. S3B). Fitted on the inferior half, it
316 predicted the superior half to 1.34, and the other way round to 0.75: the model's
317 spatial distribution of brain noise is too flat from bottom to top.

318 The per-channel pattern was poorly predicted (Fig. S3A, C). The Pearson cor-
319 relation of the log variances, model against measured, was 0.14 for the magne-
320 tometers and 0.43 for the gradiometers, against a split-half reliability of the mea-
321 surement of 0.93 and 0.97. The per-channel model/measured variance spanned
322 0.13–0.78 for the magnetometers (5th to 95th percentile) and 0.19–1.90 for the
323 gradiometers, where an exact model would give about 0.95–1.06. The correlation
324 pattern between channels agreed better: the measured and modeled correlation
325 matrices correlated at $r = 0.88$ (magnetometers) and 0.76 (gradiometers).

326 Against inter-sensor distance, the modeled magnetometer correlations
327 followed the measured ones at 25–50 mm (0.85 against 0.90) and fell faster
328 beyond (Fig. S3D). At 50–75 mm the model gave 0.59 against 0.74 measured, at
329 100–125 mm 0.20 against 0.46, and at 200–225 mm –0.30 against 0.00. Spatially
330 correlated backgrounds (correlation lengths of 5 and 10 mm) did not change this,
331 because the measured long-range correlation is a far-field pattern, not a cortical
332 one. The gradiometers' median signed correlations agreed at every distance (0.36
333 against 0.34 at 25–50 mm; Fig. S3E).

334 The measured magnetometer brain noise is more concentrated than the model's
335 (Fig. S3G, H). Its leading five components held 82 % of the variance against 72 %
336 in the model (participation ratio 5.5 against 7.4), whereas the gradiometer spectra
337 agreed (49 % against 48 %). The magnetometers' excess over the model was itself
338 concentrated: its leading three components held 91 % of it. Of its leading com-
339 ponent, 91 % lay in the 8-dimensional far-field subspace of the room-field terms,
340 against 7.8 % for a random direction.

341 A cardiac-locked average over 282 beats (heart rate 62 per minute) added
342 127 fT RMS to the median magnetometer's baseline windows. This is 25 % of
343 the measured magnetometer brain variance and 51 % of the model's shortfall, and
344 its leading component lies 91 % in the far-field subspace. With the heart's field

345 removed, the magnetometer prediction was 0.89 of the measured amplitude, and
 346 the correlation of the log variances rose to 0.76 (magnetometers) and 0.66 (gra-
 347 diometers). The heart therefore explains about half of the shortfall and much of
 348 the pattern error. The rest is concentrated in far-field components, consistent with
 349 distant or non-cortical sources, or with cortical activity correlated over longer dis-
 350 tances than the 5 and 10 mm tested.

351 With the background scale refitted per band on the gradiometers, the magne-
 352 tometer prediction fell with frequency (Fig. S4C): 0.97 at 1–4 Hz, 0.93 at 4–8 Hz,
 353 0.85 at 8–13 Hz, 0.72 at 13–20 Hz, 0.57 at 20–30 Hz and 0.33 at 30–40 Hz. The
 354 measured magnetometer excess over the gradiometer-calibrated model grew from
 355 1.07 to 9.28 in variance over the same bands. The band variances close on the
 356 whole band: the six sub-bands sum to 0.98 (magnetometers) and 0.95 (gradiome-
 357 ters) of the 1–40 Hz baseline variance. The magnetometers’ empty-room level was
 358 predicted per band to 0.75 at 1–4 Hz and 0.98–1.01 from 8–13 Hz up.

359 Detectability with a sample covariance is biased upward by the finite number
 360 of samples. The bias, computed on the surrogates, was 1.03 for the 305 channels,
 361 and every ratio below is divided by it target by target. Uncorrected, Neuromag’s
 362 detectability with the measured covariance was 1.18 [1.12, 1.24] times its modeled
 363 value; corrected, it was 1.14 [1.08, 1.20], where an exact model measured the same
 364 way gives about 1. The magnetometers alone gave 1.059 [1.003, 1.119] and the
 365 gradiometers 1.12 [1.07, 1.18]. With the heart’s field removed, the ratio was 1.15.
 366 With the measured empty room and the modeled brain noise it was 1.01, so the
 367 departure comes from the brain-noise part, not from the room.

368 By lobe, the measured-over-modeled ratio was 1.40 for the cingulate, 1.25
 369 frontal, 1.06 insula, 1.07 occipital, 1.19 parietal and 0.89 temporal. By depth,
 370 the median ratio was 1.19 at 10–15 mm, lowest (1.09) at 40–45 mm and highest
 371 (1.42) at 60–65 mm. The real noise is thus more favorable to Neuromag than the
 372 modeled noise, most for deep and cingulate sources. This is because the measured
 373 excess is concentrated in a few far-field components, which the whitening removes
 374 at little cost.

375 *S3.5. Scenarios A–E: what the check implies for the OPM*

376 The OPM’s covariance cannot be measured here. We therefore computed the
 377 implied OPM/Neuromag ratio with sensor, brain and room noise under five stated
 378 assumptions about the OPM’s real noise. The comparison is like for like with the
 379 measured covariance, which holds the room field, and uses Neuromag’s 305 good
 380 channels. In scenario A, the model’s error is common to both systems: the OPM’s
 381 real covariance changes its detectability by the same factor, target by target, as
 382 the measured covariance changes Neuromag’s, so the ratio is the model’s own. In
 383 scenario B, both systems’ cortical backgrounds are scaled by the measured mag-
 384 netometer variance excess, $k = 1.86$, with the model’s structure kept.

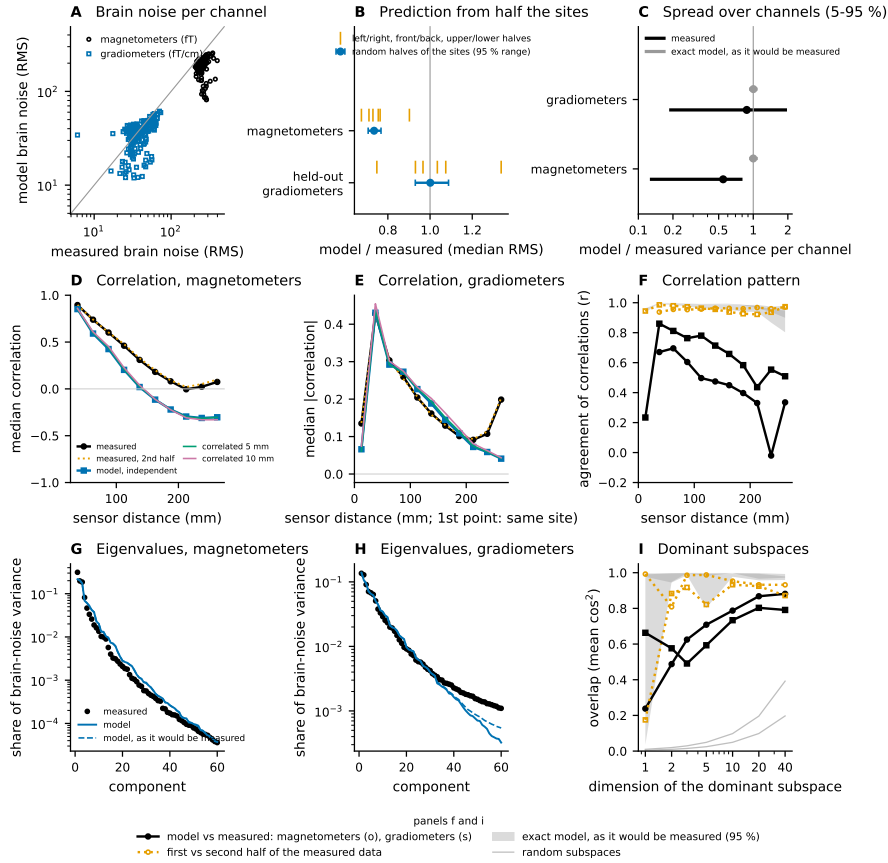


Fig. S3. Structure of the measured and modeled Neuromag brain noise (task baseline minus empty room, 1–40 Hz, the sample recording). (A) Modeled against measured brain noise per channel (RMS; magnetometers in fT, gradiometers in fT/cm, logarithmic axes; the diagonal is equality). (B) The model’s median amplitude over the measured one when the background scale is fitted on half the sites: the random halves (1,000 splits, dot and 95 % range) and the six directional halves (ticks: left and right, front and back, upper and lower), for the magnetometers and for the held-out gradiometers. (C) The 5th to 95th percentile over channels of the modeled-to-measured variance per channel (black) against the range an exact model would show when measured the same way (gray). (D) Median correlation between magnetometers against their distance: measured (both halves of the recording), the model with independent background sources and the model with correlation lengths of 5 and 10 mm. (E) The same for the gradiometers as the median absolute correlation; the first point is the two gradiometers of one site. (F) Agreement of the correlation patterns (Pearson r over the channel pairs of each distance bin) between model and measurement (black; circles, magnetometers; squares, gradiometers) and between the two halves of the measurement (orange, dotted), with the exact-model band (gray). (G, H) Eigenvalue spectra of the brain-noise covariance as shares of the variance: measured (dots), model (line) and the model as it would be measured (dashed). (I) Overlap (mean squared cosine) between the dominant subspaces of model and measurement against the subspace dimension, with the two halves of the measurement (orange), the exact-model band (gray) and random subspaces (thin gray lines). RMS, root-mean-square.

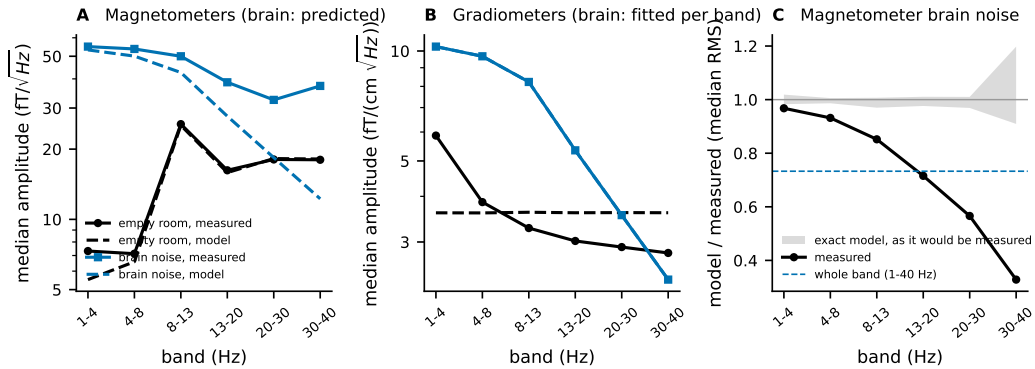


Fig. S4. The prediction by sub-band (1–4, 4–8, 8–13, 13–20, 20–30 and 30–40 Hz; the model’s cortical background rescaled in each band on the gradiometers). (A) Magnetometers: median amplitude spectral density of the measured and modeled empty room (black) and of the measured and modeled brain noise (blue), the brain noise being the prediction. (B) Gradiometers: the measured brain noise and the model fitted to it per band (blue), and the measured empty room against the model’s band-independent sensor noise (black). (C) The magnetometer brain-noise prediction per band, model over measured (median RMS), falling from 0.97 at 1–4 Hz to 0.33 at 30–40 Hz. Dashed, the whole-band value 0.73; gray, the range an exact model would give when measured the same way. RMS, root-mean-square.

Scenario C is the most pessimistic: Neuromag has its measured covariance, and the OPM bears the whole magnetometer variance excess as cortical noise and none of its measured structure. In scenario D, Neuromag has its measured covariance and the OPM is exactly as modeled, so the model’s error is specific to Neuromag. In scenario E, Neuromag has its measured empty room and the modeled brain noise and the OPM is as modeled; this isolates the sensor and room part of Neuromag’s noise.

Table S3 gives the ratios. Scenarios A, B and E left the primary model’s ratios nearly unchanged. Scenario D took the dense array to 1.034 [0.986, 1.083], no established advantage, and the site-matched array below 1. By depth, scenario D left the dense array’s median at 1.35 and 1.20 in the 10–15 and 15–20 mm bins and took it below 1 from the 25–30 mm bin (point estimates over the per-target values, room field included). Scenario C put both arrays well below 1. Which scenario holds depends on how the OPM’s real noise departs from the model, which no measurement here constrains; the noise model is validated in part for Neuromag, not for the OPM arrays.

S3.6. Sensitivity analyses

Each analysis changed one element of the adult model and recalibrated the cortical background on the measured gradiometer brain noise, as the adult compari-

Table S3. The implied OPM/Neuromag ratio under scenarios A–E. Median over the 7,661 targets; sensor, brain and room noise, unprojected; Neuromag’s 305 good channels; 95 % parcel-bootstrap intervals in brackets for selected cells; $k = 1.86$.

Assumption	Dense OPM array	Site-matched OPM array
Both systems as modeled, with sensor, brain and room noise (all 306 channels)	1.140 [1.117, 1.171]	1.007
A: the model’s error common to both systems	1.141	1.007
B: both cortical backgrounds scaled by k	1.142	1.003
C: Neuromag as measured; the OPM bears the whole excess (most pessimistic)	0.799 [0.760, 0.842]	0.675
D: Neuromag as measured; the OPM as modeled	1.034 [0.986, 1.083] (uncorrected 1.001)	0.888 [0.855, 0.921]
E: Neuromag’s measured empty room with the modeled brain noise	1.130	1.000

k , the measured magnetometer variance excess over the model. Uncorrected, before division by the finite-sample bias of the measured covariance (Section S3.4).

son does; the joint runs changed them together. Intervals are from 1,000 parcel-bootstrap resamples of one anatomy. The spike simulations were not repeated under these alternatives.

The primary model omits the cortex within 4 mm of the inner skull: 27,059 white-surface vertices, 8.7 % of the valid ones and 145 cm² of cortex (7.1 %). The lead fields of sources closer than about 2 mm to the inner-skull mesh are not converged. The primary variants therefore add the cortex from 2 mm outward, as additional background noise, as additional targets (463 of the 532 near-skull targets), or both. Removing near-skull sources from the background and removing them from the targets have opposite effects, so the exclusion is not conservative by construction.

The dense ratio was 1.137 [1.117, 1.160] with the near-skull cortex as noise, 1.152 [1.127, 1.181] with it as targets and 1.144 [1.121, 1.171] with both. The site-matched ratio was 1.008, 1.009 and 1.009, against 1.144 (dense) and 1.009 (site-matched) in the primary model. On the near-skull targets themselves, the dense array was ahead by 1.40 [1.36, 1.45]. A refined single-compartment model, whose lead fields converge nearer the skull, repeated the check down to the inner skull itself with the same result (Fig. S5B).

We gave the OPM sensor noise a 1/ f component with corners at 1, 3 and 10 Hz on top of the white level, at every level of the sweep. Two detectors were compared: the band-variance detector of the primary model, which whitens spatially only, and a frequency-resolved matched filter, which also whitens in frequency, with a flat signal spectrum (the reference) or the spike-wave spectrum. Under the frequency-

427 resolved detector, the dense ratio at 15 fT/ $\sqrt{\text{Hz}}$ was 1.127 with white noise. With
428 corners at 1, 3 and 10 Hz it was 1.122, 1.111 and 1.083 (–3.9 % against white
429 noise under the same detector; Fig. S5C, D). The site-matched ratio fell from 1.005
430 (white) to 0.980 (10-Hz corner). The frequency-resolved detector with white noise
431 lies a little below the primary model’s band-variance ratio because it also whitens
432 Neuromag’s colored brain noise in frequency.

433 We added idealized cardiac and ocular sources to the primary model. The car-
434 diac source was a current dipole 200, 250 and 300 mm below the head origin or
435 a magnetic dipole at 250 mm. The ocular sources were current dipoles at the two
436 eye centers located in the sample MRI, in the head model or in an unbounded con-
437 ductor, with three orthogonal moments each. Their level was set either to the room
438 field’s level at the median magnetometer or to the level that fills the magnetome-
439 ter brain-noise shortfall, and they were either added to the cortical background or
440 recalibrated with it.

441 After the 8-term projection, a cardiac source at 250 mm left 1.0 % of its field
442 energy at the dense array, 0.6 % at the site-matched array and 1.2 % at Neuromag
443 (Fig. S5E). Within Neuromag, it left 1.0 % at the magnetometers and 52 % at the
444 gradiometers, whose response to a distant source is small to begin with. The eyes
445 left 22 %, 11 % and 11 %. With sensor plus brain noise, unprojected, the ratios
446 changed by –1.6 % to +1.6 % for the dense array and –1.3 % to –0.2 % for the site-
447 matched array over all calibrations (Fig. S5F), because every array sees the added
448 far field in a similar proportion to its brain noise. After the projection they changed
449 by –0.9 % to +1.9 % and –1.6 % to +0.2 %.

450 The joint runs placed the near-skull cortex in the background, leaving out its
451 targets, which favor the OPM, and added a 10-Hz corner under the frequency-
452 resolved detector. The dense ratio at 15 fT/ $\sqrt{\text{Hz}}$ was then 1.080 [1.058, 1.106].
453 With the heart and eyes added at the level that fills the magnetometer shortfall it
454 was 1.091 [1.064, 1.122] (site-matched 0.980 and 0.965; Fig. S5A).

455 Over the 40 configurations of the sensitivity analysis (Table S4), the dense ratio
456 with sensor plus brain noise spanned 1.262–1.315 at 7 fT/ $\sqrt{\text{Hz}}$ and 1.072–1.152 at
457 15 fT/ $\sqrt{\text{Hz}}$. The lowest value at 15 fT/ $\sqrt{\text{Hz}}$ came from the band-variance detector
458 with the 10-Hz corner, so the joint run above is not the lowest configuration at every
459 level. At 30 fT/ $\sqrt{\text{Hz}}$ the dense ratio spanned 0.930–1.008, and the site-matched
460 ratio spanned 0.965–1.009 at 15 fT/ $\sqrt{\text{Hz}}$.

461 With the primary model, the dense ratio fell from 1.298 at 7 fT/ $\sqrt{\text{Hz}}$ through
462 1.144 at 15 fT/ $\sqrt{\text{Hz}}$ to 1.008 at 30 fT/ $\sqrt{\text{Hz}}$, and the site-matched ratio from 1.084
463 to 0.933. We found the white level at which the median ratio equals 1 on 33 log-
464 spaced levels over 4–60 fT/ $\sqrt{\text{Hz}}$, for every comparator in both conditions named in
465 advance (Table S4).

Table S4. Break-even OPM white-noise levels. The level at which the median ratio OPM/comparator equals 1, with 95 % parcel-bootstrap intervals in brackets, with sensor plus brain noise and after the 8-term projection. The last two rows give dimensionless ratios: the range of the dense and site-matched ratios over the 40 configurations of the sensitivity analysis (the primary model at five white levels, the near-skull, colored-noise and combined heart-and-eyes variants and the joint runs), in both conditions.

Array and comparator	Sensor plus brain noise	Projected
<i>Break-even level ($fT/\sqrt{Hz}$)</i>		
Dense OPM against Neuromag (306 channels)	31.5 [28.5, 34.9]	26.0 [22.3, 29.5]
Dense OPM against the magnetometers	37.4 [34.0, 41.2]	33.1 [30.0, 36.9]
Dense OPM against the gradiometers	54.2 [51.9, 56.9]	39.2 [33.9, 43.3]
Site-matched OPM against Neuromag (306 channels)	16.7 [14.0, 19.4]	8.9 [5.6, 12.4]
Site-matched OPM against the magnetometers	21.6 [17.8, 24.7]	14.8 [11.1, 18.8]
Site-matched OPM against the gradiometers	34.2 [32.8, 36.1]	17.8 [12.8, 22.7]
<i>Range of the ratio over the configurations, at 7 / 15 / 30 $fT/\sqrt{Hz}$</i>		
Dense OPM against Neuromag	1.262–1.315 / 1.072–1.152 / 0.930–1.008	1.242–1.300 / 1.044–1.136 / 0.855–0.971
Site-matched OPM against Neuromag	1.076–1.123 / 0.965–1.009 / 0.820–0.933	1.019–1.061 / 0.890–0.950 / 0.706–0.833

The heart or the eyes alone change the dense ratio by -1.6% to $+1.6\%$ (Section S3.6).

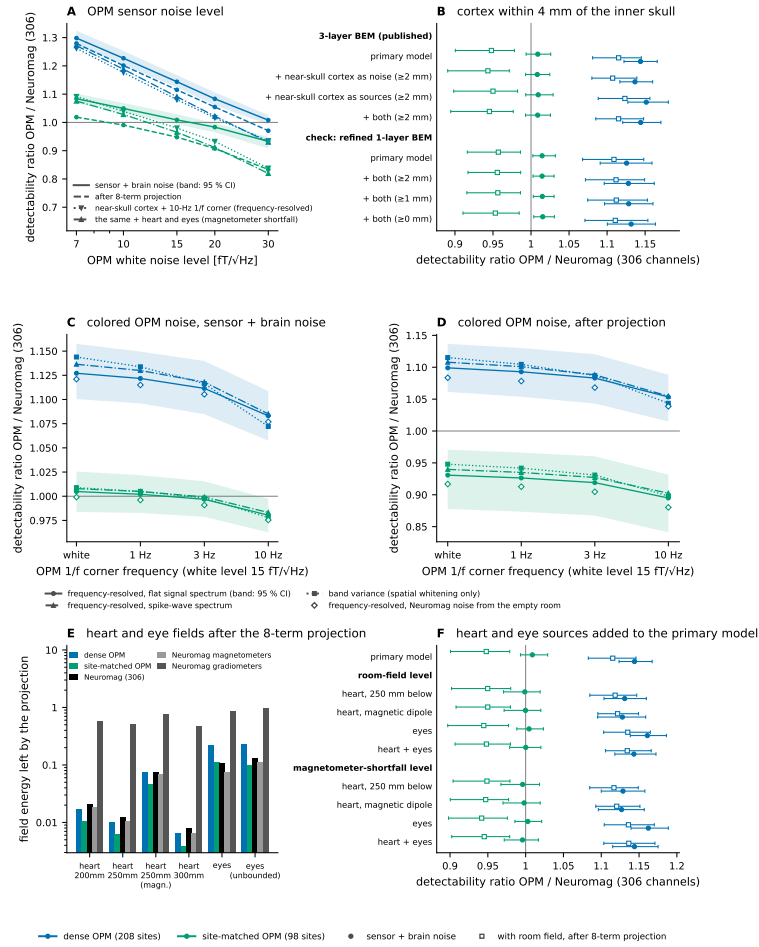


Fig. S5. Sensitivity of the adult comparison to the noise model's alternatives. Median over the 7,661 targets of the paired ratio OPM/Neuromag (306 channels) for the dense OPM array (208 sites, blue) and the site-matched OPM array (98 sites, green), with 95 % parcel-bootstrap bands or bars. Filled circles (B, F), sensor plus brain noise; open squares (B, F), with the room field, after the 8-term projection. (A) The ratio against the OPM white-noise level (7, 10, 15, 20 and 30 fT/√Hz): the primary model with sensor plus brain noise (solid, band) and after the projection (dashed); the near-skull cortex with a 10-Hz 1/f corner under the frequency-resolved detector (dotted, inverted triangles); and the same with the heart and eyes at the magnetometer-shortfall level (dash-dotted, triangles). (B) The cortex within 4 mm of the inner skull added to the primary three-layer model as noise, as sources or both (from 2 mm outward), and the same check with a refined single-compartment model down to the inner skull. (C, D) Colored OPM noise at 15 fT/√Hz against the 1/f corner frequency, with sensor plus brain noise (C) and after the projection (D): the frequency-resolved detector with a flat signal spectrum (circles, band) and with the spike-wave spectrum (triangles), the band-variance detector of the primary model (squares, dotted) and the frequency-resolved detector with Neuromag's noise taken from the empty-room recording (open diamonds). (E) The share of a far-field source's field energy left after the projection, for cardiac dipoles 200, 250 and 300 mm below the head origin, a magnetic dipole at 250 mm and ocular dipoles in the head model or an unbounded conductor, at the dense array, the site-matched array, Neuromag's 306 channels, its magnetometers and its gradiometers (logarithmic axis). (F) The ratio with these sources added to the primary model at the room field's level and at the magnetometer-shortfall level.

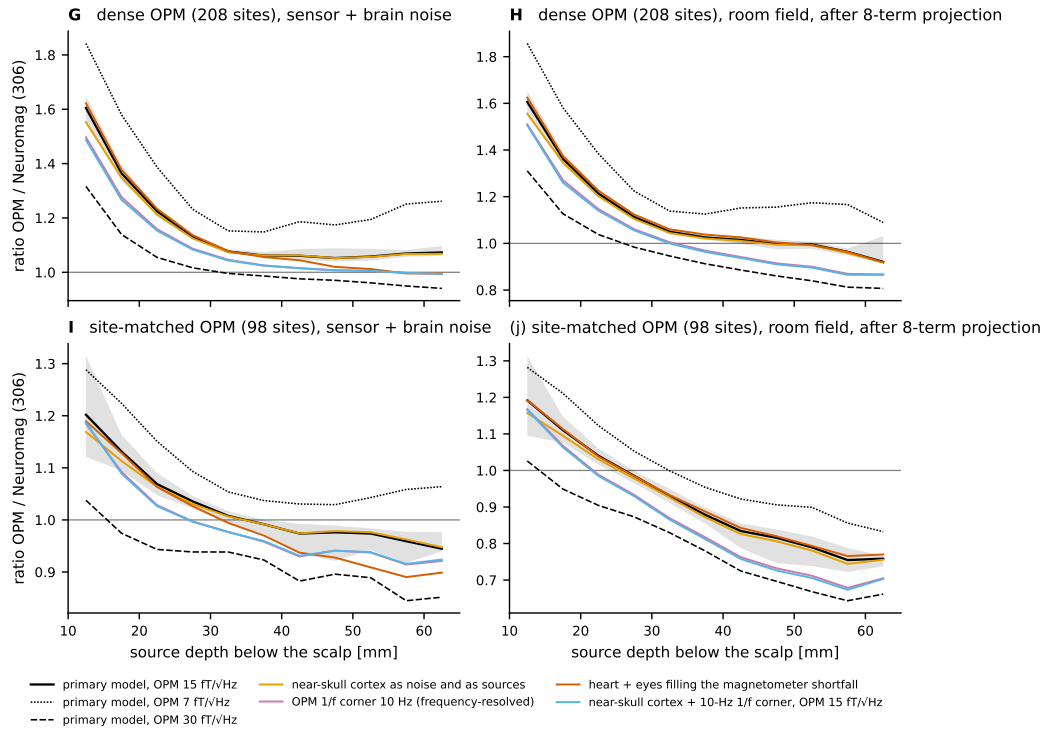


Fig. S5. (Continued.) (G–J) The ratio against depth below the scalp (5-mm bins) for the dense (G, H) and site-matched (I, J) arrays, with sensor plus brain noise (G, I) and after the projection (H, J). Lines show the primary model at 15 fT/√Hz (black, with its band), at 7 fT/√Hz (dotted) and at 30 fT/√Hz (dashed); the near-skull cortex as noise and as sources; the 10-Hz corner under the frequency-resolved detector; the heart and eyes filling the magnetometer shortfall; and the near-skull cortex with the 10-Hz corner at 15 fT/√Hz.

466 **S4. Head models and the MRI check**

467 *S4.1. The adult and the three classes of smaller head*

468 The adult is the MNE sample subject (Gramfort et al., 2013), with FreeSurfer
469 surfaces and atlas labels (Desikan et al., 2006; Fischl, 2012) and a three-layer
470 boundary-element model (BEM) whose outer surface lies on the magnetic reso-
471 nance imaging (MRI) scalp. Its usable cortex is 1,878 cm². The smaller heads
472 belong to three classes, which are not independent subjects (main text, Section
473 2.2).

474 The two scaled adults are copies of the adult scaled by one factor, so they differ
475 from it in size only. The school-age size control is 0.895 times the adult, and the
476 2-year size control has the head circumference of the 24-month template. Both
477 keep the adult’s vertices, so Δ is taken vertex by vertex.

478 The three infant templates at 24, 18 and 12 months (O’Reilly et al., 2021)
479 average many MRIs of the Neurodevelopmental MRI Database (Richards et al.,
480 2016) and have smooth cortices. Their usable cortex is 1,062, 975 and 895 cm²,
481 48–57 % of the adult’s, which lowers their total background power and their patch
482 cancellation. They were used at their native size with their own three-layer BEMs,
483 oct-6 source spaces, atlas labels and fiducials.

484 The three school-aged children A, B and C (7.8–8.7 years; subjects sub-Z213,
485 sub-Z209 and sub-Z226) are typically developing children of the OpenNeuro
486 dataset ds005234 (Fadeev et al., 2024, 2025), which is de-identified and released
487 under a CC0 license. Their own MRIs were processed by the dataset’s authors.
488 We used their white and sphere surfaces, atlas annotations and dense MRI
489 scalp as distributed; these share each child’s volume information. The dataset’s
490 boundary-element surfaces were also taken from the file tree as distributed, in
491 which each subject’s watershed surfaces sit in the preceding subject’s folder. We
492 matched these files to each child by the volume information they share with the
493 child’s own surfaces, and the storage objects behind them carry the name of the
494 subject they belong to.

495 These watershed surfaces put the inner skull implausibly close to the scalp, a
496 median 0.8–2.5 mm below it over the upper head, with an outer skull crossing it.
497 We therefore modeled the children’s skull. The inner skull was moved inward to 8
498 mm below the scalp wherever the dataset’s lay shallower, with no vertex within 2
499 mm of a white-surface vertex. The surface between vertices comes within 0.9–1.3
500 mm of the white surface, and no white vertex lies outside it. The outer skull was
501 placed about halfway between the inner skull and the scalp. Over the upper head,
502 the modeled inner skull lies a median 5.7–7.8 mm below the scalp. On the adult,
503 whose segmented skull is known, the same procedure applied to a degraded inner
504 skull changed the primary ratio by at most 0.003 over the modeled skull depths.

505 As for every head, the head surface of the forward model is the watershed outer
506 skin placed on the MRI scalp.

507 The dataset provides no fiducials. The children’s fiducials define the head
508 frame and hence the centered placement, the origin of every scaled helmet and
509 the coverage region of the optically pumped magnetometer (OPM) arrays. They
510 were transferred from the adult by a similarity fit of the cortices and moved to the
511 nearest scalp vertex. On the templates, which have their own fiducials, this trans-
512 fer reproduced the fiducials to within 1.6–14.4 mm and the head frames to within
513 3.2–5.9°. The largest error, 14.4 mm, was at the nasion of the 18-month template,
514 which lies farthest from the cortex.

515 Frame rotations of this size (total angles, their axis not resolved) are of the
516 order of the rotations of the placement family (pitch $\pm 10^\circ$, yaw $\pm 10^\circ$, roll $\pm 5^\circ$;
517 Section S5.1). This comparison indicates rather than bounds their effect on the
518 children’s placements, and the transferred fiducials were not checked on the chil-
519 dren’s own MRIs. We did not use fiducials derived from Montreal Neurological
520 Institute (MNI) coordinates: on the adult, such fiducials land 10.0–27.4 mm from
521 the digitized ones.

522 Every head uses the adult’s conductivities (0.3, 0.006 and 0.3 S/m). A single-
523 compartment BEM (0.3 S/m inside the inner skull) needs no skull or scalp con-
524 ductivity. With it, the difference of medians from the adult was +0.19 to +1.13
525 dB, against +0.19 to +1.14 dB with three layers. Across all smaller heads, that
526 difference changed by -0.06 to $+0.01$ dB.

527 As distributed, the skull of the 12-month template is 0.25 mm thick at its
528 thinnest: 23 of its 2,562 inner-skull vertices lie within 1 mm of the outer skull.
529 The single-compartment BEM has no skull and is unaffected by this. With it, the
530 template’s D was +2.03 dB, against +2.04 dB with three layers, and its difference
531 from the adult was +1.06 dB, against +1.05 dB. The thin skull therefore does not
532 carry the template’s result.

533 *S4.2. Regions*

534 Regions are the lobes of the Desikan–Killiany atlas (Desikan et al., 2006),
535 grouped as follows. Frontal: superior, rostral middle and caudal middle frontal,
536 pars opercularis, triangularis and orbitalis, lateral and medial orbitofrontal, precen-
537 tral, paracentral and frontal pole. Parietal: superior and inferior parietal, supra-
538 marginal, postcentral and precuneus. Temporal: superior, middle and inferior
539 temporal, banks of the superior temporal sulcus, fusiform, transverse temporal,
540 entorhinal, temporal pole and parahippocampal. Occipital: lateral occipital, lin-
541 gual, cuneus and pericalcarine. Cingulate: rostral and caudal anterior, posterior
542 and isthmus. The sixth region is the insula. The medial wall belongs to no region,
543 and both hemispheres are pooled.

544 We also report single parcels that the epilepsy literature emphasizes: pre-
545 central, superior temporal (“lateral temporal” in the figures) and parahippocam-
546 pal, with a mesial temporal group of the parahippocampal and entorhinal parcels;
547 “mesial temporal” always means this group. All sources lie on the cortical surface,
548 and the hippocampus and amygdala are not modeled. The parahippocampal and
549 entorhinal cortex, 26.8–28.7 mm below the scalp in the adult (median over targets),
550 are therefore the nearest proxies for mesial temporal generators. The regions were
551 chosen from the literature, and rankings among them are exploratory.

552 *S4.3. The MRI quality check of the school-aged children and the infant templates*

553 The children’s white surfaces come within 3.7–5.7 mm of the scalp surface
554 used for sensor placement and source depths. Child B has 247 cortical targets
555 shallower than 10 mm, and the adult has none. Child B’s head is the size of the
556 24-month template’s (head circumference 48.6 against 49.5 cm; cap volume 1,795
557 against 1,802 cm³), while its cortex is much larger (1,666 against 1,062 cm²). Both
558 observations pointed to a scalp surface lying inside the true scalp, or misregistered
559 relative to the white surface. We therefore ran an MRI quality check on the three
560 children, with the adult and the 24-month template as references, using the surfaces
561 exactly as the forward models use them. The same check was then run on the 18-
562 and 12-month templates.

563 Along the outward normal of every vertex of the dense scalp, the T1 intensity
564 was sampled from –20 to 20 mm in 0.25-mm steps. The MRI head boundary is
565 where this profile falls through the volume’s air/tissue threshold (Otsu’s method;
566 the primary edge), with the half-maximum edge and the steepest gradient as sec-
567 ondary definitions. The signed offset of this boundary from the scalp used, positive
568 outward, was summarized over the cap (the head above the plane of the fiducials)
569 by its median, quartiles and 5th to 95th percentiles. It was also decomposed into a
570 uniform offset, a translation and a rotation.

571 The frames were checked by comparing the volume geometry of every surface
572 with the T1 header. The registration of the white surface to the T1 was checked by
573 the white/gray contrast across it: the rigid shift that maximizes this contrast was
574 searched coarsely and then finely, and we report the contrast gain of the best shift.
575 The head circumference was measured both on the MRI head boundary and on the
576 scalp used. Targets below 8 and 10 mm were counted against both surfaces.

577 A head is classed as misregistered if a surface’s header differs from the T1
578 by more than the header tolerance, or if the white surface’s realigning shift, edge
579 translation or rotation displacement exceeds 1 mm. It is classed as inside or outside
580 if the cap median of the boundary’s offset exceeds, or falls short of, the adult’s
581 by more than 1 mm (one voxel) at both the Otsu and the half-maximum edge.
582 Otherwise it is classed as usable.

583 We tested the check on the adult. A declared inward displacement of the adult's
584 scalp by 4 mm was recovered as 4.0 mm, and a 2-mm translation as 2.03 mm, with
585 97 % of the offset pattern explained by a translation. A 2-mm shift of the adult's
586 white surface raised the T1 contrast at the surface by 65 % when the shift was
587 undone, so a misregistered white surface would be found.

588 Every surface's header matched its T1, with corner displacements of at most
589 2.7×10^{-5} mm in the children (Fig. S6). The children's white surfaces were regis-
590 tered: the best realigning shift was 0.0 mm, the edge translation at most 0.01 mm
591 and the rotation displacement at most 0.02 mm. The contrast gain was 0 %, 0 %
592 and 0 % for children A, B and C.

593 In the children, the scalp used lay a median 1.3–1.5 mm inside the MRI head
594 boundary at the Otsu edge (0.4–1.2 mm at the half-maximum edge), against 1.3
595 mm and 1.1 mm in the adult (Fig. S6C). A FreeSurfer scalp thus sits just inside the
596 T1 edge in every individual head, the adult's included. The scalp of the 24-month
597 template lay 1.6 mm outside its boundary (verdict: outside), which is consistent
598 with an average template whose scalp is smoothed outward. The scalps of the 18-
599 and 12-month templates lay 1.6 and 1.3 mm outside theirs (2.3 and 1.6 mm at the
600 half-maximum edge, against 1.6 mm for the 24-month template).

601 The verdict for the 12-month template was outside, and for the 18-month tem-
602 plate misregistered. The white surface of the 18-month template fitted the T1's
603 white/gray contrast best after a rigid shift of 2.3 mm, which raised the contrast by
604 41 %, against 0.5 mm in the other two templates. The T1 edge across it showed a
605 translation of 1.7 mm; both values exceed the 1-mm tolerance. This is a fit over
606 the whole surface, which single sections do not show; the template's white-surface
607 vertex closest to the scalp used lies 7.3 mm below it. Whether the misfit reflects a
608 displaced surface or the weaker contrast of an average image was not settled. No
609 template surface was repaired, and an offset of that size was not propagated to the
610 template's results.

611 The head circumference on the MRI boundary minus that on the scalp used was
612 +8.9 to +10.7 mm in the children (49.7–54.5 against 48.6–53.5 cm) and +9.9 mm
613 in the adult (59.6 against 58.6 cm). Measured from the MRI boundary instead of
614 the scalp used, the median target depth changed by +1.3 to +1.5 mm in the children,
615 +1.3 mm in the adult and –1.7 to –1.4 mm in the templates, whose scalps lie outside
616 their boundaries. The children's medians thus shifted as the adult's did, but their
617 shallowest targets did not, as follows.

618 The shallow cortex is not an artifact of scalp placement or registration. The
619 closest white-surface vertex lay 3.7, 3.7 and 5.7 mm below the scalp used in chil-
620 dren A, B and C (4.6–7.5 mm below the MRI boundary), against 9.7 mm in the
621 adult and 9.2 mm in the 24-month template (Fig. S6A). The 5th percentile of the
622 white surface's depth was 7.9–12.0 mm in the children, against 14.5 mm in the
623 adult and 11.8 mm in that template. This is a comparison with the adult; no pedi-

624 atric norm is applied. In no child does the T1 head mask reach a face of the volume
625 other than the inferior one. It reaches that face at the neck in children B and C, as
626 in the adult, so the field of view does not cut the head above the neck.

627 Within 8 mm of the scalp used lay 22.4, 83.4 and 3.6 cm² of white surface in
628 children A, B and C, and within 8 mm of the MRI boundary 6.8, 28.8 and 0.1 cm²
629 (adult: none; Fig. S6B). Of the cortical targets, 3, 18 and 0 lay below 8 mm, and
630 43, 247 and 11 below 10 mm (adult: 0; counted from the full-precision depths).
631 Measured to the MRI head boundary instead of the scalp used, child B's targets
632 less than 10 mm deep fell from 247 to 47. The verdicts were child A usable, child B
633 usable and child C usable (3 of 3 usable). No surface was repaired, and no analysis
634 needed rerunning.

635 The children's shallow cortex, and child B's small head around a near-adult
636 cortex, are therefore not explained by their scalp surfaces or registration. One
637 explanation the data leave open is a head boundary that the T1 images themselves
638 place too deep. Such a boundary would bring the FreeSurfer scalp, which follows
639 the image's edge, toward the cortex together with the dataset's watershed surfaces
640 (inner skull a median 0.8–2.5 mm below the scalp; Section S4.1), and a check
641 against the same images cannot detect it. An independent measurement of the
642 scalp and skull over the upper head, from another sequence or modality, would
643 test it.

644 The check cannot settle whether the white surface is anatomically correct
645 within a few millimeters of the scalp, where the modeled scalp-and-skull layer is
646 thinner than the adult's, nor whether three children of one dataset represent their
647 age. The verdicts test registration and the scalp's position against the T1. They do
648 not test anatomical plausibility, such as tissue thicknesses against norms or head
649 circumference against centiles for age.

650 The source rule (no source within 4 mm of the inner skull) left out 7.3 %,
651 10.0 % and 3.3 % of the white-surface area of children A, B and C, against 7.2 %
652 in the adult. The adult's share is the 7.1 % of Section S3.6 plus the little cortex
653 outside the inner skull, 0.05 cm² of it within 10 mm of the scalp. The rule left out
654 9.3–10.6 % in the scaled adults and 0.8–2.6 % in the templates. In the children, the
655 excluded area includes 94–100 % of the cortex within 8 mm of the scalp: of child
656 B's 14,761 white-surface vertices there, 570 pass the source rule. For these rea-
657 sons, the main text treats the children's results as provisional geometric examples.

658 **S5. Helmet constructions**

659 *S5.1. The fixed adult helmet and its placements*

660 We used the same Neuromag superconducting quantum interference device
661 (SQUID) sensors and the same noise for every head. The centered placement
662 puts each head's own head frame where the adult's lies in the sample recording

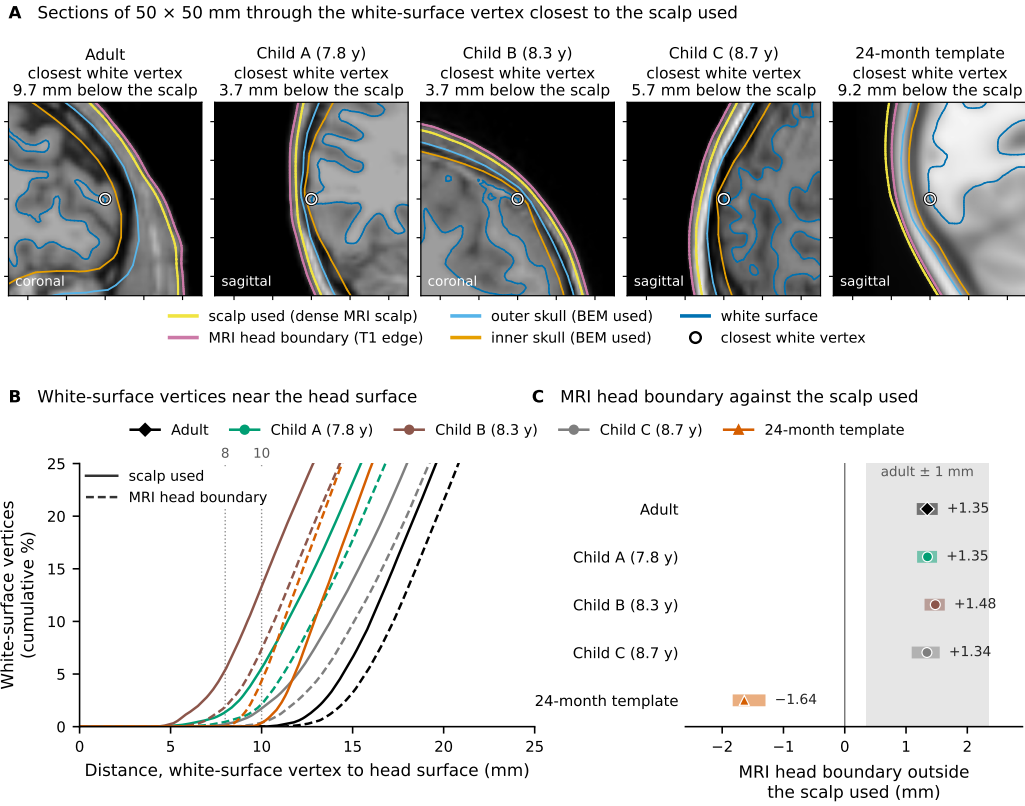


Fig. S6. MRI quality check of the school-aged children's anatomy, with the adult and the 24-month template as references. The surfaces are those the forward models use. (A) Sections of 50 × 50 mm through the white-surface vertex closest to the scalp used (open circle; 9.7 mm below the scalp in the adult, 3.7, 3.7 and 5.7 mm in children A, B and C, and 9.2 mm in the 24-month template), sagittal or coronal, whichever contains more of the scalp's outward normal at that vertex. Gray: T1 intensity; yellow: the dense MRI scalp used for sensor placement and source depths; pink: the MRI head boundary, where the T1 intensity falls through the volume's air/tissue (Otsu) level along the scalp's outward normals; light blue and orange: the outer and inner skull of the BEM used (the children's inner skull modeled 8 mm below the scalp where the dataset's lay shallower); dark blue: the white surface. (B) Cumulative share of white-surface vertices against their distance to the scalp used (solid) and to the MRI head boundary (dashed), lowest quarter, with the check's thresholds at 8 and 10 mm (dotted). (C) Offset of the MRI head boundary from the scalp used over the head above the fiducial plane: median and quartiles, positive where the boundary lies outside the scalp; shaded: the adult's median ± 1 mm, the tolerance of the verdict. The children's scalps lie a median 1.3–1.5 mm inside their MRI head boundary, as the adult's does (1.3 mm), and the 24-month template's lies 1.6 mm outside it. Children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025); template: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016); adult: MNE sample data. BEM, boundary-element model; MRI, magnetic resonance imaging.

663 (the recording's device-to-head transform), so that the ear line stays where the
664 adult's was. The primary placement is top contact: the head is raised from the
665 centered pose until the nearest magnetometer coil center is 20 mm from the scalp,
666 the 18-mm Dewar spacing plus a margin. This is a best-fit assumption rather than
667 documented practice. A child in an adult helmet may be placed centrally and repo-
668 sitioned toward the expected source (Gaetz et al., 2008), and a small child's shoul-
669 ders can stop the head short of the top (Tesan et al., 2010). Top contact therefore
670 favors Neuromag, and the other placements bound this choice.

671 A family of 12 placements chosen from the geometry alone (source-blind) tests
672 the sensitivity to placement; for child C, one lateral shift was infeasible, leaving
673 11. The family comprises top contact, back contact, shifts of ± 5 mm left-right
674 and front-back, and rotations of $\pm 10^\circ$ pitch, $\pm 5^\circ$ roll and $\pm 10^\circ$ yaw, each raised to
675 the same contact. Three further placements are the centered pose itself, a laterally
676 centered pose raised to contact (the head moved sideways to the helmet's midline
677 before the raise) and true contact at the Dewar spacing.

678 The adult's own D depends on the placement: +1.27 dB at its measured pose,
679 +1.00 dB at top contact, +1.01 dB laterally centered at contact and +0.87 dB at
680 true contact. Over its 12 source-blind placements, the adult's D spanned +0.98 to
681 +1.43 dB (a spread of 0.46 dB), with top contact at rank 2 from the lowest. In the
682 fixed helmet, each smaller head was compared with the adult placed by the same
683 rule. Their Δ was +0.76 to +2.50 dB at the centered pose, +0.17 to +1.07 dB at
684 top contact, +0.09 to +1.06 dB laterally centered at contact and +0.17 to +1.09 dB
685 at true contact.

686 With sensor plus brain noise, the point estimates had the same sign against the
687 gradiometers (+0.11 to +1.49 dB), against the magnetometers (+0.25 to +0.98 dB)
688 and for the site-matched array (+0.08 to +1.10 dB). With the room field projected
689 out, Δ was unchanged in the scaled adults (+0.44 to +1.09 dB, against +0.44 to
690 +1.07 dB) and lower in the templates (+0.42 to +0.83 dB, against +0.73 to +0.96
691 dB). In the children it was small: +0.03 [-0.17, +0.22], +0.26 [+0.04, +0.39] and
692 -0.19 [-0.38, +0.18] dB for children A, B and C, child C's point estimate negative
693 with an interval spanning zero. The site-matched arrays gave -0.06, -0.17 and
694 -0.10 dB, every interval including 0.

695 The adult's placement spread exceeded Δ in 4 of the 8 smaller heads. Each
696 child's own D varied by 0.24, 0.57 and 0.86 dB (A, B and C) over the same family.
697 Against each head's family median instead of top contact, the difference of pooled
698 medians from the adult shrank by 0.05–0.20 dB.

699 The optically pumped magnetometer (OPM) arrays were refitted to each head
700 with the adult's rules: 144–174 dense and 80–91 site-matched sites on the smaller
701 heads, at a median standoff of 6.99–7.00 mm above every head's own magnetic
702 resonance imaging (MRI) scalp. They are the same in every helmet. Sensor noise,
703 room field and analysis band were unchanged. The cortical background variance

704 per unit cortical area was calibrated once on the adult, so a smaller cortex carries
705 less total background; as a sensitivity analysis, this variance was multiplied by 0.5
706 and by 2.

707 In the realistic heads, fit and head size can act on D through at least four routes.
708 A wider SQUID gap and a shallower cortex would raise it, whereas fewer OPM
709 sites and, in the school-aged children, a near-adult cortex close to the OPMs would
710 lower it. The constructions below vary these routes together in different combina-
711 tions, and none isolates one route (main text, Section 3.5). The last route rests on
712 the children's unverified near-scalp anatomy (Section S4.3).

713 *S5.2. The helmet scaled with the head*

714 Every coil center is scaled about the head origin by the child-to-adult ratio of
715 head circumferences, keeping coil size, orientation and noise. The factor is raised
716 in steps until no coil is within 18 mm of the scalp. The helmet is thus scaled
717 in proportion to the head, gaps included, unlike the constant gap of the sphere
718 (Section S2). It was computed about the centered head and, as a variant, about
719 the laterally centered head; this variant was added after the first pediatric results
720 were known, to remove the templates' lateral offset. The construction asks what
721 the adult's fit would give at a child's size; it is not a model of a pediatric SQUID
722 device. The main text treats it as a secondary construction, because it also brings
723 the sensors closer than the adult's own fit (see below).

724 In the helmet scaled about the laterally centered head, Δ was -0.76 to $+0.04$
725 dB against the adult in its own scaled helmet. The highest upper interval bound,
726 child C's, was $+0.20$ dB (parcel-bootstrap intervals, which exclude placement and
727 construction uncertainty). Only child C, whose scaled helmet keeps a median gap
728 of 33.1 mm against the adult's 29.6 mm, was not negative ($+0.04$ [-0.20 , $+0.20$]
729 dB, an interval that also contains its fixed-helmet Δ). With the room field projected
730 out, the difference of medians from the adult in its own scaled helmet was -0.92
731 to -0.10 dB (-0.67 to $+0.05$ dB with sensor plus brain noise).

732 Scaled about the centered head instead, the helmet left residual gains in the
733 heads that sit off-center: the 24- and 18-month templates and child C ($+0.24$, $+0.18$
734 and $+0.56$ dB; Δ -0.74 to $+0.56$ dB overall). The OPM's absolute advantage re-
735 mained in the scaled helmet (D $+0.62$ to $+1.33$ dB in the smaller heads), but the
736 additional child-adult contrast is no longer established.

737 Three checks qualify this result. First, part of the drop comes from the ref-
738 erence: against the adult at top contact rather than in its own scaled helmet, the
739 difference of medians was -0.38 to $+0.33$ dB (child C $+0.33$ dB, above its fixed-
740 helmet Δ). Second, scaling the helmet also scales the gap, which shrank below the
741 adult's 29.6 mm in 7 of the 8 smaller heads (25.1–28.0 mm). This helmet therefore
742 fits most smaller heads more tightly than the adult's helmet fits the adult, which
743 favors Neuromag; the helmet fitted at the adult's gap (Section S5.3) avoids this

744 confound. Third, the smaller heads carry fewer OPM sites (144–174 against 208).
 745 The adult’s array subsampled to those counts lost 0.27–0.53 dB, so at an equal
 746 OPM site count Δ in the fixed helmet would be +0.43 to +1.43 dB.

747 *S5.3. The helmet fitted at the adult’s gap*

748 The Neuromag helmet is scaled about the head origin of each head’s laterally
 749 centered pose until the median gap from magnetometer coil center to scalp equals
 750 the adult’s. We used two targets: 29.6 mm, the adult’s median gap in its own helmet
 751 about its laterally centered head (scale factor 1.000), and, as a variant, 28.4 mm,
 752 the adult’s gap at top contact in the fixed helmet. Coil size, orientation, integration
 753 points and noise are kept. No coil center may come within the 18-mm Dewar
 754 spacing of the scalp or lie inside the head. In child C, this clearance binds before
 755 the target is reached: the helmet stops at a gap of 32.8 mm, 3.3 mm above the
 756 target, with a scale factor of 0.941 instead of the 0.920 that the target would need.

757 The OPM arrays are those of the fixed-helmet analysis, unchanged. A within-
 758 head contrast, the median over a head’s targets of the target-wise difference be-
 759 tween D at top contact in the fixed helmet and D in the fitted helmet, is therefore a
 760 property of Neuromag alone and is the same whichever OPM array is compared.
 761 Table S5 gives the gaps and scale factors of every helmet per head. Intervals use
 762 1,000 parcel-bootstrap resamples for the dense array in the fixed helmet and in the
 763 helmet fitted at 29.6 mm, and 200 elsewhere. This helmet is a geometric control
 764 at the adult’s median gap, not at its gap distribution, placement or coverage, and it
 765 is not a pediatric SQUID system.

766 Table S6 gives, per head, Δ at the adult’s gap for the dense and site-matched
 767 arrays, Δ with the room field projected out, Δ at the adult’s top-contact gap, the
 768 within-head contrast, the interaction and Δ at equal OPM site count. At the adult’s
 769 gap, Δ was +0.02 to +0.03 dB in the scaled adults, +0.04 to +0.15 dB in the tem-
 770 plates and –0.39 to +0.01 dB in the children. The intervals included zero in 5 heads
 771 (2-year size, 24-month template, 18-month template, 12-month template and child
 772 C), lay below zero in 2 (child A and child B) and above zero in 1 (school-age size;
 773 its lower bound rounds to 0.00 from above). The site-matched array gave –0.45
 774 to +0.13 dB, the projected condition –0.67 to –0.05 dB, and the variant at the
 775 top-contact gap –0.36 to +0.13 dB.

776 Against the adult at top contact in the fixed helmet, rather than in its fitted
 777 helmet, the heads fitted at the adult’s top-contact gap differed from it by +0.10 to
 778 +0.23 dB in the scaled adults and templates (4 of 5 intervals above zero). In the
 779 children they differed by –0.11 [–0.40, +0.06], –0.22 [–0.35, +0.01] and +0.30
 780 [–0.02, +0.61] dB. This residual depends on how the adult is placed. The interac-
 781 tion (below), which places the adult by the same rule in both helmets, was +0.40 to
 782 +1.10 dB in the scaled adults and templates at the adult’s gap and +0.40 to +1.11
 783 dB at its top-contact gap. The dense array kept its absolute advantage in every head

Table S5. The helmets per head. Median gap from magnetometer coil center to scalp in the fixed adult helmet at top contact, in the helmet scaled with the head about the laterally centered head, and in the helmet fitted at the adult's gap (29.6 mm) with its scale factor k ; and k for the variant fitted at the adult's top-contact gap (28.4 mm).

Head	Gap, fixed helmet (mm)	Gap, scaled with the head (mm)	Gap, fitted at the adult's gap (mm)	k , fitted	k , fitted at the top-contact gap
Adult	28.4	29.6	29.6	1.000	0.991
School-age size	34.1	26.6	29.6	0.916	0.908
2-year size	39.1	25.1	29.6	0.876	0.868
24-month template	38.5	27.0	29.6	0.881	0.873
18-month template	38.9	27.2	29.6	0.880	0.872
12-month template	40.4	27.7	29.6	0.855	0.846
Child A	35.2	26.1	29.6	0.921	0.913
Child B	38.4	28.0	29.6	0.877	0.867
Child C	32.6	33.1	32.8 ^a	0.941	0.941

^a Clearance-bound: the helmet stops at the 18-mm Dewar spacing before the target gap is reached. Gap: median over the magnetometer coil centers of the distance to the nearest scalp point; k : scale factor of the helmet about the head origin.

784 in the fitted helmet (D +0.97 to +1.32 dB over the nine heads). The site-matched
785 array's D in the fitted helmet was −0.09 to +0.33 dB with sensor plus brain noise
786 and −0.96 to −0.10 dB projected.

Table S6. The helmet fitted at the adult’s gap, per head. Dense OPM array against Neuromag (306 channels) with sensor plus brain noise unless stated; values in dB; brackets: 95 % parcel-bootstrap intervals within the head.

Head	Δ , dense (dB)	Δ , site-matched (dB)	Δ , dense, projected (dB)	Δ , dense, at the top-contact gap (dB)	Within-head contrast (dB)	Interaction (non-additivity) (dB)	Δ at equal OPM site count (dB)
School-age size	+0.03 [0.00, +0.05]	+0.03	−0.05	+0.02	+0.17 [+0.08, +0.28]	+0.40 [+0.30, +0.50] (+0.02)	+0.20 [+0.16, +0.24]
2-year size	+0.02 [−0.02, +0.06]	+0.12	−0.07	+0.01	+0.76 [+0.64, +0.90]	+1.10 [+0.97, +1.21] (−0.05)	+0.30 [+0.24, +0.35]
24-month template	+0.04 [−0.06, +0.19]	+0.02	−0.25	+0.02	+0.34 [+0.20, +0.54]	+0.54 [+0.44, +0.73] (+0.15)	+0.42 [+0.29, +0.68]
18-month template	+0.15 [−0.08, +0.33]	−0.02	−0.21	+0.13	+0.42 [+0.27, +0.61]	+0.70 [+0.45, +0.83] (+0.03)	+0.42 [+0.22, +0.69]
12-month template	+0.05 [−0.09, +0.24]	−0.07	−0.25	+0.02	+0.55 [+0.38, +0.78]	+0.66 [+0.56, +0.99] (+0.24)	+0.49 [+0.33, +0.81]
Child A	−0.30 [−0.48, −0.22]	−0.26	−0.67	−0.27	+0.29 [+0.19, +0.40]	+0.53 [+0.44, +0.70] (+0.08)	+0.16 [−0.12, +0.23]
Child B	−0.39 [−0.54, −0.17]	−0.45	−0.50	−0.36	+0.42 [+0.27, +0.61]	+0.71 [+0.58, +0.77] (+0.09)	+0.16 [−0.10, +0.41]
Child C	+0.01 [−0.23, +0.18]	+0.13	−0.27	+0.11	−0.12 [−0.20, −0.02]	+0.04 [−0.11, +0.19] (+0.13)	+0.48 [+0.14, +0.62]

Δ : against the adult in its own fitted helmet ($D = +1.28$ dB). Projected: room field projected out. Within-head contrast: the area-weighted median over the head’s targets of the target-wise difference, D in the fixed helmet at top contact minus D in the fitted helmet (the adult’s own is -0.23 dB). Interaction: the difference of the two heads’ within-head contrasts, taken like Δ (per vertex for the scaled adults, per parcel for the templates and children) and summarized by its area-weighted median; it is therefore not the difference of the medians in the previous column. In parentheses: the non-additivity, $\Delta(\text{fixed}) - \Delta(\text{fitted}) - \text{interaction}$, of the area-weighted medians. Δ at equal OPM site count: Δ with the adult’s dense array subsampled to the head’s dense site count. OPM, optically pumped magnetometer.

787 The within-head contrast measures what Neuromag loses, in a given head, to
788 the fit, gap and placement of the fixed helmet together. It was +0.17 to +0.76 dB
789 in the scaled adults, +0.34 to +0.55 dB in the templates and -0.12 to +0.42 dB in
790 the children. The adult's own contrast was -0.23 dB, because its laterally centered
791 pose in its own helmet is looser than top contact. Child C's contrast was negative
792 although its fixed-helmet gap (32.6 mm) and its clearance-bound fitted gap (32.8
793 mm) nearly coincide. The fitted construction itself is worth -0.11 [-0.15, -0.08]
794 dB in the adult at an equal median gap, so this contrast is one of construction, not
795 of gap.

796 The interaction, the smaller head's contrast minus the adult's, is the extra that
797 the fixed helmet gives a smaller head beyond what it gives the adult. It was +0.40
798 to +1.10 dB in the scaled adults, +0.54 to +0.70 dB in the templates and +0.04 to
799 +0.71 dB in the children. At every vertex of a scaled adult, the interaction equals
800 $\Delta(\text{fixed}) - \Delta(\text{fitted})$, because the same vertex is compared in both helmets. The
801 area-weighted medians of these quantities are not additive, so $\Delta(\text{fixed}) - \Delta(\text{fitted})$
802 - interaction leaves a residual (Table S6). The within-head and interaction es-
803 timates are therefore reported in their own right rather than derived from the Δ
804 values.

805 To equalize OPM site counts, the adult's dense array was subsampled by
806 farthest-point sampling to each smaller head's dense site count (144–174 sites),
807 with the adult in the same fitted helmet. At the adult's gap, Δ was then +0.16
808 to +0.49 dB: +0.20 to +0.30 dB in the scaled adults, +0.42 to +0.49 dB in the
809 templates and +0.16 to +0.48 dB in the children (the intervals of children A and
810 B include zero). At the adult's median gap and equal OPM site count, a smaller
811 head's dense array thus gains a little. One possibility, which these controls do not
812 test, is that its cortex lies closer to the OPMs. In the scaled adults and templates,
813 the fewer sites take about that much back. In children A and B, whose intervals
814 at equal OPM site count include zero, the fewer sites take back more, and what
815 else lowers their Δ is not settled by these controls (the children are provisional
816 examples; Sections S4.3 and S5.5).

817 As a measure of placement uncertainty, Δ in the fixed helmet was computed
818 at each of the 12 source-blind placements, each head against the adult at the same
819 placement, as a difference of the area-weighted medians (Table S7). In the scaled
820 adults, the band was +0.20 to +0.67 dB at its lowest and +1.12 to +1.61 dB at its
821 highest; in the children, -0.10 to +0.19 dB at its lowest and +0.37 to +0.70 dB at
822 its highest. It was positive at every placement in every head except child C. The
823 crossed band, which pairs any feasible placement of the head with any placement
824 of the adult, is wider and includes zero in the children. Top contact was not an
825 extreme of any band (Table S7, rank).

Table S7. Placement bands of Δ in the fixed adult helmet. Dense OPM array against Neuromag (306 channels), sensor plus brain noise.

Head	n	Band (dB)	Span (dB)	Rank of top contact	Crossed band (dB)	Δ at top contact (dB)
School-age size	12	+0.20 to +1.12	0.92	9	+0.03 to +1.58	+0.44 [+0.34, +0.56]
2-year size	12	+0.67 to +1.61	0.94	8	+0.26 to +2.07	+1.07 [+0.91, +1.18]
24-month template	12	+0.48 to +1.38	0.90	7	+0.32 to +1.53	+0.73 [+0.46, +0.98]
18-month template	12	+0.54 to +1.36	0.82	11	+0.37 to +1.51	+0.88 [+0.54, +1.05]
12-month template	12	+0.75 to +1.66	0.92	10	+0.55 to +1.82	+0.96 [+0.61, +1.51]
Child A	12	+0.06 to +0.37	0.30	10	-0.18 to +0.52	+0.30 [+0.11, +0.38]
Child B	12	+0.19 to +0.63	0.44	9	-0.03 to +0.99	+0.41 [+0.28, +0.54]
Child C	11	-0.10 to +0.70	0.80	5	-0.38 to +0.94	+0.17 [-0.07, +0.34]

Band: Δ at each of the source-blind placements, each head against the adult at the same placement, as a difference of area-weighted medians; n : feasible placements; rank: the rank of top contact from the lowest Δ of the band; crossed band: any feasible placement of the head against any placement of the adult. Last column: the primary Δ at top contact with its 95 % parcel-bootstrap interval, for comparison. OPM, optically pumped magnetometer.

826 *S5.4. Depth matching*

827 Depth is the route of head size that the sphere describes (Section S2). The cor-
828 tex of the templates and children is shallower than the adult's: area-weighted me-
829 dian depths were 20.6–23.2 mm (templates) and 20.0–23.4 mm (children), against
830 26.2 mm. Of the cortical area, 36–46 % (templates) and 35–47 % (children) lay
831 10–20 mm below the scalp, against 21 % in the adult. In the fixed helmet, reweight-
832 ing each head's targets to the adult's depth distribution lowered the difference of
833 pooled medians from the adult. In the templates it fell from +0.85 to +1.05 dB
834 pooled to +0.29 to +0.53 dB reweighted, and in the children from +0.19 to +0.52
835 dB to –0.05 to +0.05 dB.

836 Within depth strata, where Δ is the difference of the two heads' area-weighted
837 medians, Δ was positive in every stratum of the scaled adults and templates (+0.11
838 to +1.62 dB; Fig. S7A). For the individual children, it was negative in 11 of the 12
839 child-strata at 10–30 mm depth (–0.36 to +0.08 dB). No interval there lay above
840 zero, the largest lower bound being –0.25 dB, and Δ was positive deeper (+0.01
841 to +1.77 dB; Fig. S7B). In this decomposition, the children's fixed-helmet gain is
842 matched by their shallower cortex. The depths are measured from the scalp used,
843 which is unverified near the scalp in the children (Section S4.3). Measured from
844 the MRI boundary instead, they would shift by +1.3 to +1.5 mm in the children,
845 +1.3 mm in the adult and –1.7 to –1.4 mm in the templates. The decomposition
846 describes the model's surfaces, not a tested mechanism.

847 In the helmet fitted at the adult's gap, the children's strata were negative
848 throughout the depths that hold most of their cortex. For child A, Δ at the adult's
849 gap was –1.70 dB at 10–15 mm, –0.91 dB at 20–25 mm and –0.20 dB at 30–40
850 mm, and its intervals included zero only from the 40–50 mm stratum down. The
851 two decompositions, fitting the helmet and matching depth, are not additive, and
852 neither apportions the routes exactly. The fitted helmet is the main control because
853 it matches the adult's median gap, although it changes the gap distribution, the
854 placement and the coverage with it.

855 *S5.5. Noise in the smaller heads and the share of cortex where the OPM is ahead*

856 The noise differs between the classes of head under the convention that the
857 background variance per unit cortical area is the adult's. In the fixed helmet, Neu-
858 romag's magnetometer brain noise fell from 202 fT in the adult at top contact (192
859 fT at its measured pose) to 98–150 fT in the templates and scaled adults. Both their
860 wider gap and their smaller cortex contribute, and their shares were not separated:
861 moving the school-age size control from its centered pose up to top contact raised
862 this noise from 131 to 150 fT. A smaller cortex with the same background per unit
863 area does not lower the field at sensors that stay close, as the OPMs show (525 and
864 537 fT in the scaled adults, against 494 fT in the adult).

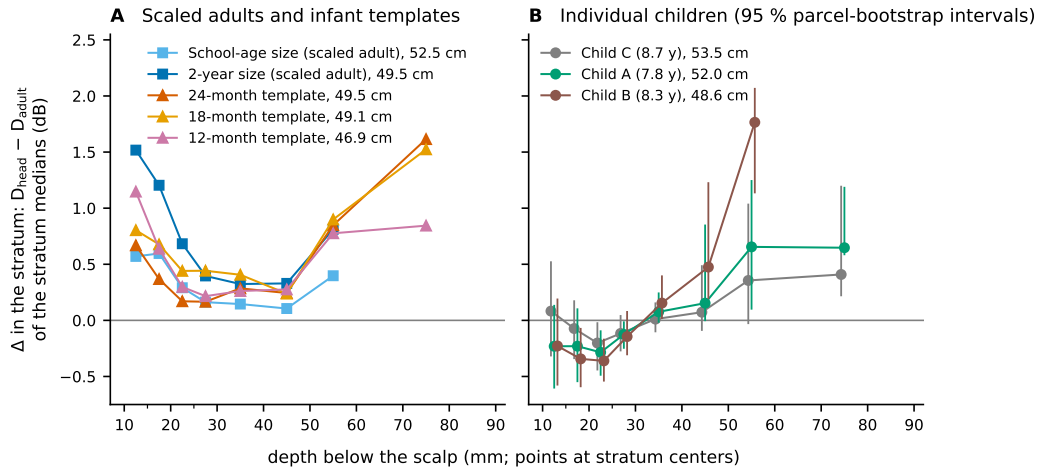


Fig. S7. Δ within depth strata in the fixed adult helmet. Δ within strata of depth below each head's own scalp, as the difference of the two heads' area-weighted medians in the stratum (points at the stratum centers). (A) Scaled adults and infant templates. (B) School-aged children, with 95 % intervals from parcels resampled in each anatomy independently (1,000 resamples). Strata with fewer than 10 targets in either head are omitted. Fixed adult helmet at top contact, dense optically pumped magnetometer (OPM) array against Neuromag (306 channels), sensor plus brain noise.

865 The children's cortex is nearly adult-sized (1,666–1,869 cm²). Neuromag's
 866 brain noise therefore stayed at 168–194 fT in the children. The OPMs, close to a
 867 near-adult cortex that lies near the scalp in these surfaces (unverified there; Sec-
 868 tion S4.3), saw more of it (617–751 fT).

869 The share of cortex on which the OPM is ahead, the kind of metric used by Jas
 870 et al. (2026), is saturated for the dense array (99.6 % of cortical area in the adult).
 871 For the site-matched array, it was 49 % in the adult at top contact (against 56 %
 872 of all targets at the measured pose) and 58–89 % in the smaller heads in the fixed
 873 helmet (Fig. S8C).

874 S6. Regions and the noise floor across heads

875 This section gives the regional results for every head and the dependence of
 876 the comparison on the noise of the optically pumped magnetometers (OPMs).
 877 Throughout, D is the detectability of the dense OPM array relative to Neuromag,
 878 the 306-channel superconducting quantum interference device (SQUID) system,
 879 in dB (Section S1). D is the area-weighted median over a region's cortical targets
 880 in both hemispheres, with sensor plus brain noise. The helmet constructions are
 881 described in Section S5.

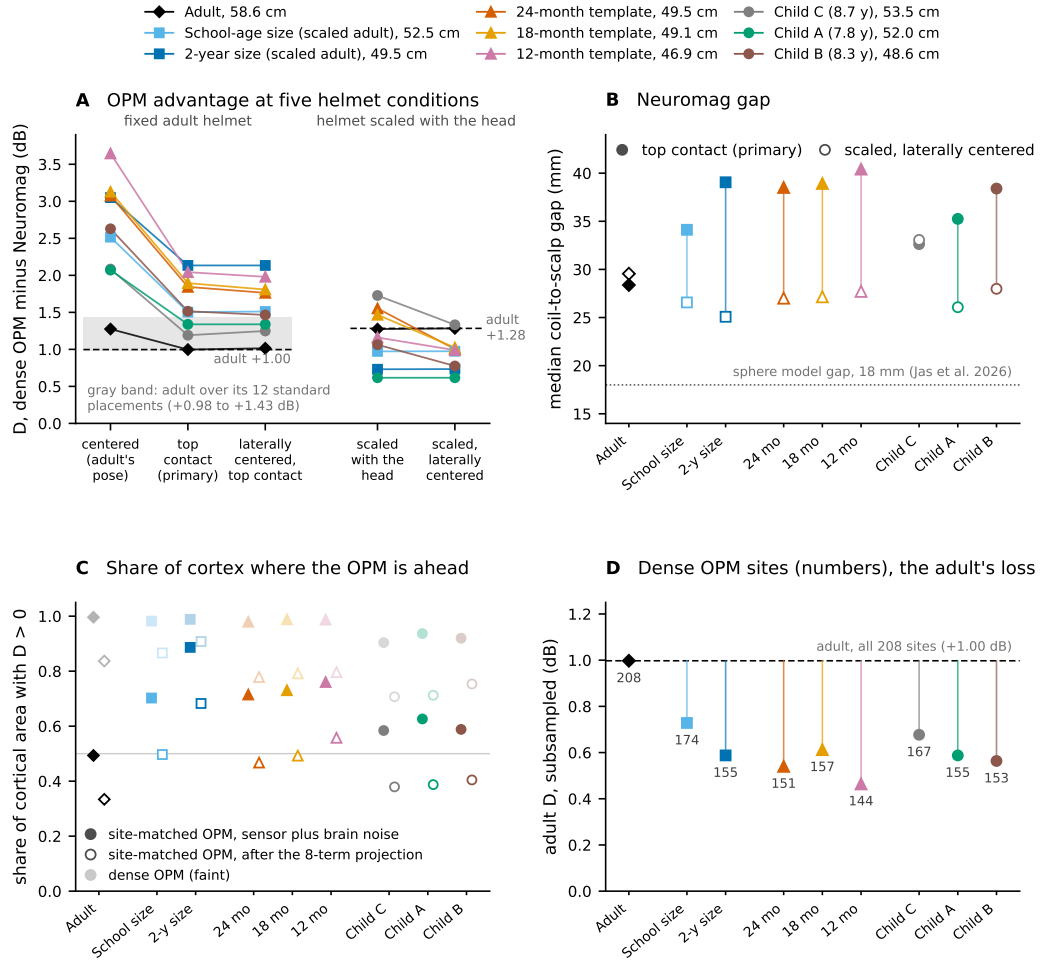


Fig. S8. The helmet scaled with the head and the routes of head size. Dense OPM array against Neuromag (306 channels), sensor plus brain noise; D is the area-weighted median over cortical targets; heads as in the legend, ordered by class and head circumference. (A) D under five helmet conditions: the fixed adult helmet at the centered pose, at top contact (primary) and laterally centered at top contact, and the helmet scaled with the head about the centered and about the laterally centered head. Dashed lines: the adult at top contact (+1.00 dB) and in its own scaled, laterally centered helmet (+1.28 dB); gray band: the adult over its 12 source-blind placements (+0.98 to +1.43 dB). (B) Median gap from magnetometer coil to scalp at top contact (filled) and in the scaled, laterally centered helmet (open); dotted: the 18-mm gap of the spherical benchmark (Jas et al., 2026). (C) Share of cortical area with $D > 0$ in the fixed adult helmet at top contact, for the site-matched array (filled: sensor plus brain noise; open: room field projected out) and the dense array (faint). (D) Dense OPM sites per head (numbers) and the adult's D with its array subsampled to each count, from the adult's +1.00 dB with all 208 sites. The helmet fitted at the adult's gap is shown in the main text (Fig. 7). OPM, optically pumped magnetometer.

882 *S6.1. Regions by head*

883 In the fixed adult helmet at top contact, the dense array was ahead in every
884 region of every head (D +0.25 to +3.06 dB over the cells of Fig. S9A). A smaller
885 head exceeded the adult's value for the same region in 72 of 80 cells. Of the six
886 lobes, the temporal lobe gained most in the infant templates (+2.63, +2.88 and
887 +3.02 dB at 24, 18 and 12 months, against +1.17 dB in the adult).

888 The parahippocampal and mesial temporal rows were the adult's lowest tempo-
889 ral rows (+0.74 and +0.74 dB), and both were higher in every template and scaled
890 adult. The mesial temporal group reached +1.16 to +2.79 dB in the templates and
891 scaled adults but +0.46 to +1.77 dB in the children, with child C (+0.46 dB) below
892 the adult. The cingulate was the lowest row in every head (+0.25 dB in the adult,
893 +0.26 to +0.61 dB in the smaller heads).

894 The peak-channel signal-to-noise ratio (SNR) gave a different picture (sensor
895 plus brain noise). Against the magnetometers alone, the dense array's peak-channel
896 SNR was ahead in every head (D +1.84 dB in the adult at top contact, +2.05 to
897 +2.71 dB in the smaller heads). Against Neuromag's best channel, Neuromag led
898 in the adult (-1.11 dB) and in 4 of the 8 smaller heads (-0.85 to +0.79 dB). The
899 children were among them (-0.64, -0.28 and -0.85 dB in children A, B and C).

900 In the helmet scaled with the head, the regional gain was smaller, and how
901 much smaller depended on the reference (lobes only; Fig. S9B). Against the adult in
902 its own scaled helmet, 39 of the 48 lobe cells of the smaller heads were at or below
903 the adult's. Against the adult at top contact, 28 were (17 of 30 for the templates
904 and scaled adults, 11 of 18 for the children). These counts compare point estimates
905 without a tolerance. The children's lobes ranged from +0.17 to +1.62 dB in the
906 scaled helmet.

907 Table S8 gives the regional D in the helmet fitted at the adult's gap. The adult's
908 ordering, with frontal, precentral and lateral temporal cortex highest and cingulate
909 and insula lowest, held in the scaled adults. The cingulate was the lowest row in
910 every head except the children, where the insula lay just below it. The templates
911 and children, however, reordered the upper rows. The parahippocampal and mesial
912 temporal rows were high in the 24- and 18-month templates and in child B, and
913 the occipital lobe was highest in child C.

914 The children differed from the adult by region in both directions. Child B was
915 below the adult in the frontal lobe (-0.81 dB) and the precentral gyrus (-0.63 dB)
916 but above it in the parahippocampal (+0.86 dB) and occipital (+0.23 dB) cortex.
917 Child C was below the adult in the entorhinal cortex (-1.02 dB) but above it in the
918 occipital lobe (+0.64 dB).

919 The cortical maps (Fig. S10; main text, Fig. 6C) show where D is largest:
920 on the lateral convexity, most on gyral crowns, with medial surfaces near zero.
921 Region by region, then, the ordering changed with the helmet; the relative weight

Table S8. Regional D in the helmet fitted at the adult’s gap. Dense OPM array against Neuromag (306 channels), sensor plus brain noise; area-weighted median over the region’s cortical targets, both hemispheres; point estimates.

Region	D (dB)								
	Adult	School	2-y	24 mo	18 mo	12 mo	Child A	Child B	Child C
Frontal	+1.82	+1.80	+1.77	+1.89	+2.05	+2.19	+1.22	+1.01	+1.52
Parietal	+1.15	+1.34	+1.22	+1.05	+0.97	+1.02	+0.74	+0.89	+1.53
Temporal	+1.32	+1.28	+1.20	+1.40	+1.50	+1.35	+1.13	+1.21	+1.03
Occipital	+0.95	+0.93	+0.85	+0.82	+0.79	+0.74	+1.18	+1.18	+1.58
Cingulate	+0.40	+0.41	+0.39	+0.56	+0.50	+0.46	+0.30	+0.37	+0.41
Insula	+0.71	+0.66	+0.67	+1.08	+0.96	+0.74	+0.28	+0.32	+0.40
Precentral	+1.79	+1.70	+1.57	+1.56	+1.73	+1.91	+1.10	+1.16	+1.56
Superior temporal	+1.39	+1.50	+1.22	+1.09	+1.27	+1.11	+0.96	+0.71	+0.94
Parahippocampal	+0.88	+0.87	+1.01	+1.97	+2.25	+0.88	+0.81	+1.74	+1.49
Mesial temporal	+0.91	+0.82	+0.93	+1.80	+2.01	+1.10	+0.37	+1.32	+0.76

Mesial temporal: parahippocampal and entorhinal cortex together. School and 2-y: the school-age and 2-year size controls (scaled adults); 24, 18 and 12 mo: the infant templates; Child A–C: the school-aged children. D: OPM detectability relative to Neuromag; OPM: optically pumped magnetometer.

of helmet and region was not quantified. Heat maps and cortical maps show point estimates, and the pediatric intervals represent within-anatomy resampling only.

S6.2. The noise floor across heads

D fell with the OPM’s sensor noise in every head (Fig. S11). At 30 fT/√Hz, about the empty-room floor read from the recording of Jas et al. (2026), the dense array’s D was −0.05 dB in the adult and +0.27 to +0.97 dB in the smaller heads. The site-matched array’s D was negative in every head except the 2-year size control (−0.79 dB in the adult, −0.41 to +0.18 dB in the smaller heads).

The difference from the adult itself depended on the noise. At 7 fT/√Hz the individual children were slightly below the adult (−0.15 to −0.03 dB, difference of medians), and at 30 fT/√Hz they were above it (+0.32 to +0.77 dB). A plausible reading, not tested here, is that the children’s OPM channels, dominated by brain noise, lose less to a noisier sensor than the adult’s. A noisier OPM thus leaves the adult without an established advantage first, while the smaller heads in the fixed helmet keep part of theirs.

S7. The spike study: design, results and localization

S7.1. The exploratory design

The detection study ran the noise model in the time domain (Table S18). The background moments had a 1/f spectrum, the room field had the cross-spectra of

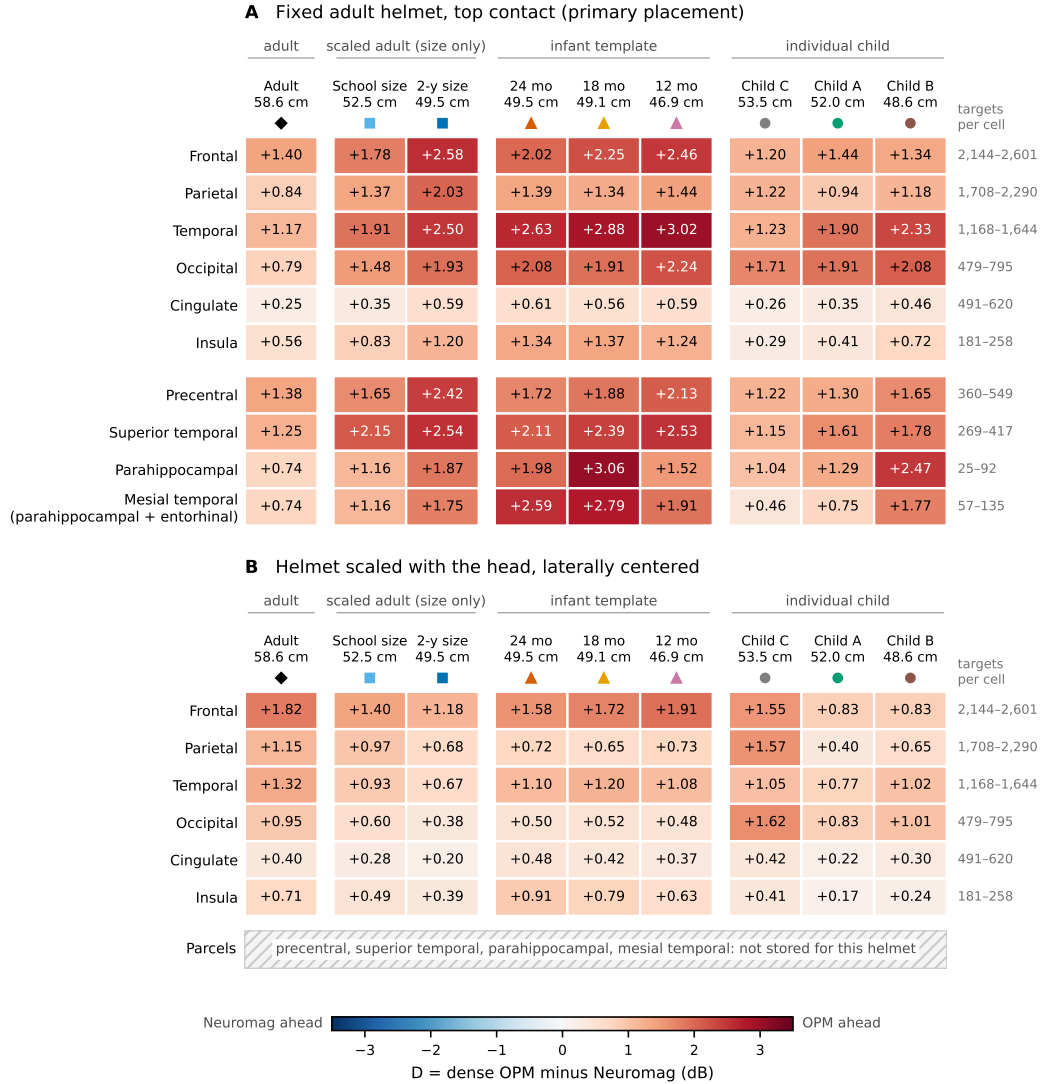


Fig. S9. Brain regions that favor the OPM at each head size. Area-weighted median D, dense OPM array minus Neuromag (306 channels), sensor plus brain noise, by lobe and for the precentral, superior temporal, parahippocampal and mesial temporal (parahippocampal and entorhinal) cortex; both hemispheres; medial wall excluded. The numbers at right give the targets per row (range over heads). Heads are grouped by class and ordered by head circumference (46.9–58.6 cm). (A) The fixed adult helmet at top contact. (B) The helmet scaled with the head, laterally centered; the parcel rows are hatched because per-target values were not computed for this helmet (the helmet fitted at the adult’s gap gives them; Table S8). The color scale is shared and centered at 0 dB; every cell is a point estimate. D: OPM detectability relative to Neuromag; OPM: optically pumped magnetometer.

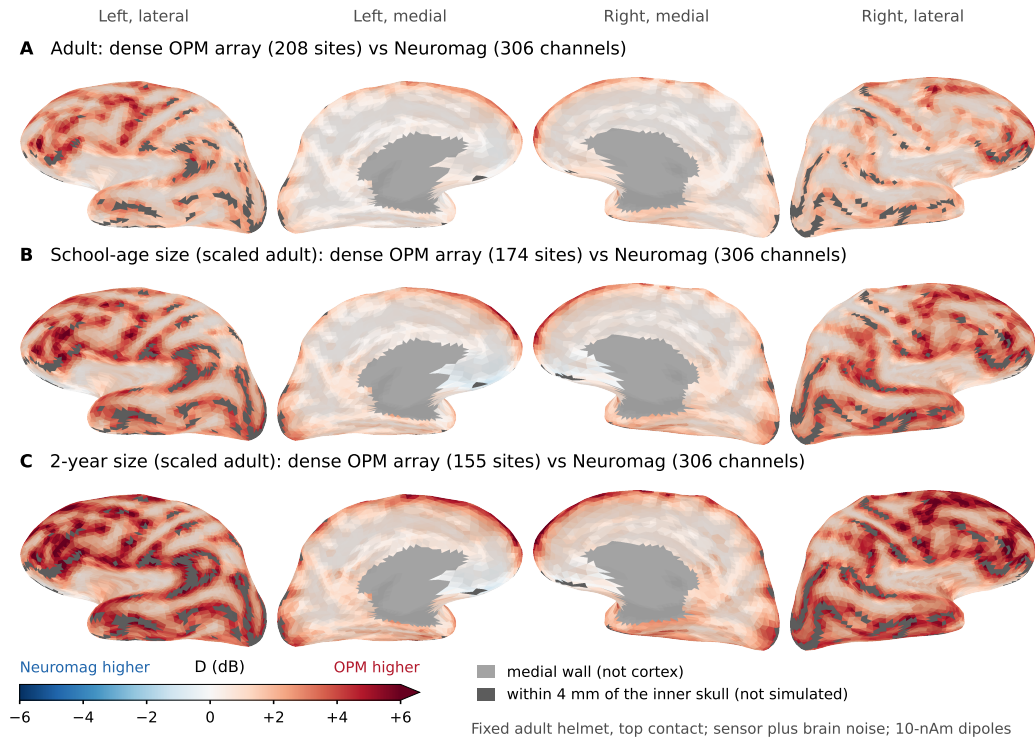


Fig. S10. D on the cortex of the adult and the scaled adults. D of the dense OPM array against Neuromag (306 channels) at top contact in the fixed adult helmet, sensor plus brain noise, on the inflated cortex, in lateral and medial views of both hemispheres. (A) The adult (208 dense sites; area-weighted median D +1.00 dB). (B) The school-age size control (174 sites; +1.50 dB). (C) The 2-year size control (155 sites; +2.13 dB). The size controls share the adult's vertices. Red: the OPM's detectability is higher; the color scale is that of the cortical maps of the main text. Light gray: medial wall; dark gray: cortex within 4 mm of the inner skull, not simulated. D: OPM detectability relative to Neuromag; OPM: optically pumped magnetometer.

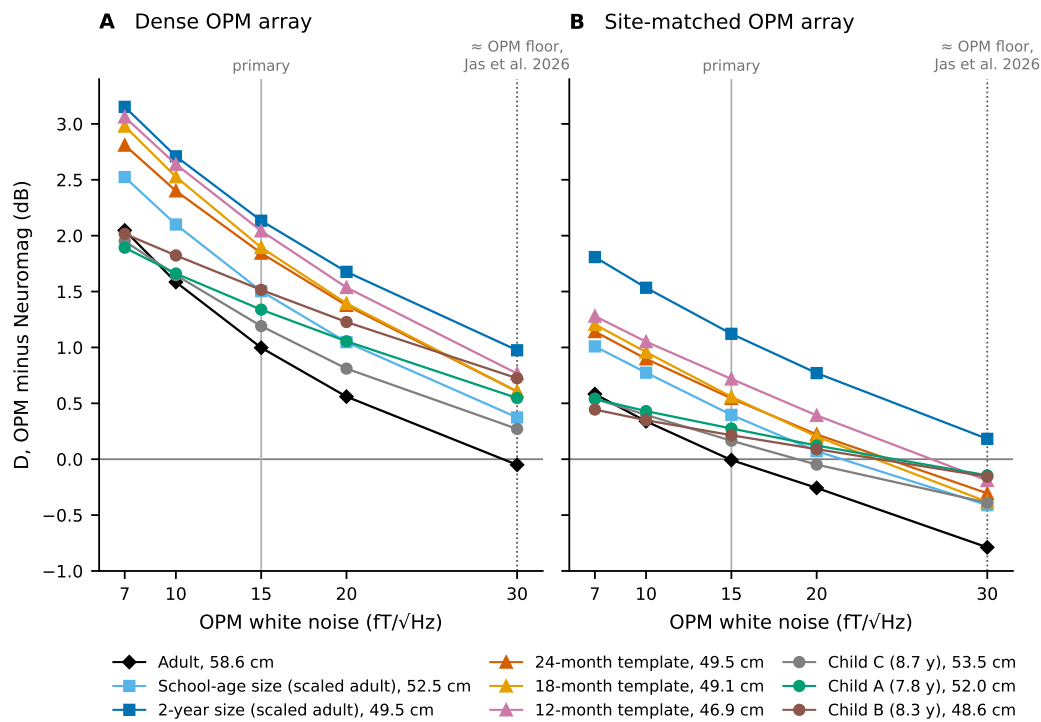


Fig. S11. The OPM noise level across heads. D (OPM minus Neuromag, 306 channels; sensor plus brain noise; fixed adult helmet at top contact; area-weighted median over cortical targets) against the OPM's white sensor noise (7, 10, 15, 20 and 30 fT/√Hz), with Neuromag's noise unchanged, for the adult and the eight smaller heads. (A) Dense OPM array. (B) Site-matched OPM array. Solid vertical line: the primary 15 fT/√Hz. Dotted line: 30 fT/√Hz, about the OPM empty-room floor read from the spectra of Jas et al. (2026), a broadband value not matched to the 1–40 Hz band. D: OPM detectability relative to Neuromag; OPM: optically pumped magnetometer.

941 the empty-room recording, and the sensor noise was white. Each component was
942 scaled so that its in-band covariance equaled that of the adult model, and the room
943 field was not projected out. The data were filtered to 1–40 Hz and decimated to
944 150.2 Hz, with one noise realization per 30-s segment shared by all arrays. The
945 spike was the spike-wave complex of Hunold et al. (2016), digitized, stretched in
946 time by 0.75, 1 and 1.5 (3 morphologies) and passed through the same filter.

947 Spikes were placed at 72 locations stratified by depth below the scalp (10–20,
948 20–30, 30–45 and 45–70 mm; 18 locations per band) and by orientation (3 bands
949 from radial to tangential; 6 locations per depth and orientation stratum). Each
950 location received 18 focal-dipole events, at 6 strengths of 10, 20, 40, 80, 160 and
951 320 nAm, each with every morphology. It also received patch events of 10 mm
952 radius, which carried the same total moments with the unstretched morphology.
953 This gave 1,296 focal and 432 patch events, 1,728 per anatomy, injected in random
954 order one every 2 s and identically in every array. The adult’s locations lay 24 in
955 the frontal lobe, 14 in the parietal, 14 in the temporal, 12 in the cingulate, 6 in the
956 insula and 2 in the occipital lobe, at median depths of 16.9, 26.0, 36.8 and 51.7
957 mm in the four bands.

958 Every array was read by two detectors, and Neuromag was also read by its mag-
959 netometers alone and by its gradiometers alone, giving 5 sensor sets per anatomy,
960 each read by both detectors. Both detectors whitened the data with a Ledoit–Wolf
961 covariance (Ledoit and Wolf, 2004) from 10 min of null data and filtered them
962 with the spike templates. The oracle detector knows the source and the time. It
963 tests the whitened, template-filtered projection on the true field pattern at the true
964 peak (one-sided) against a threshold set for a per-trial false-positive probability of
965 0.001.

966 The practical detector knows neither the source nor the time. At every sample
967 it takes the largest normalized output over the 3 templates and 441–716 candidate
968 field patterns on a 10-mm cortical grid that excludes the true sources. Its local
969 maxima at least 0.25 s apart are events. A spike counts as detected when an event
970 above threshold lies within ± 50 ms of its peak, and the same events are counted as
971 false on null data. The templates are the simulated morphologies and the candi-
972 dates’ field patterns come from the true forward model, which makes the practical
973 detector idealized for every array. Whether it is equally idealized for every array
974 was not tested in the exploratory run; the confirmatory run’s mismatch variant tests
975 one departure (Section S7.5), and focal on-scalp patterns may be more sensitive to
976 forward-model and coregistration error.

977 The practical detector’s thresholds for 1 false event per minute were set on 20
978 min of independent null data and frozen. On 20 min of held-out null data they gave
979 0.40–1.55 false events per minute across sensor sets and anatomies (Table S9). In
980 the adult, the practical detector needed 1.83–2.08 times the oracle’s strength for
981 half detection at 10–20 mm.

Table S9. Held-out false-event rates of the exploratory run. False events per minute on the 20 min of held-out null data at the frozen thresholds of the practical detector, which were set for 1 false event per minute on 20 min of calibration null data.

Head	Neuromag (306 channels)	Site-matched OPM array	Dense OPM array
Adult	1.30	0.70	0.95
School-age size	1.40	0.50	1.05
2-year size	0.40	0.65	1.20
24-month template	0.70	1.55	1.30
18-month template	0.90	1.45	1.10
12-month template	1.20	0.95	0.85
Child A	0.90	1.35	0.75
Child B	0.80	1.55	0.70
Child C	1.20	0.70	0.75

OPM, optically pumped magnetometer. The adult sits at its measured head position, the other heads at top contact in the fixed adult helmet.

S50 is the strength at which the detection rate of a band's focal events reaches 50 %, interpolated linearly in log strength between the tested strengths. A band holds 324 focal events: the morphologies pooled over the band's 18 locations. The patch events enter neither S50 nor the paired tests. The paired ratio $S50_{\text{Neuromag}}/S50_{\text{OPM}}$ is above 1 when the optically pumped magnetometer (OPM) array needs less strength (Section S1). The adult sits at its measured pose and the smaller heads at top contact. The measured pose gives the adult a detectability ratio D 0.28 dB higher than top contact does, so the adult's spike results are not placed by the same rule as those of the smaller heads.

No exploratory endpoint was named before the analyses. The endpoints, chosen after the analyses, were three. The first was the dense array's detectability ratio against Neuromag's 306 channels in the adult. The second was Δ of the dense array against the same comparator at top contact in the fixed helmet and in the helmet scaled with the head about the laterally centered head, each against the adult placed by the same rule, reported descriptively. The third, for the spikes, was the dense array against Neuromag's 306 channels in the 10–20 mm band with the practical detector. It was tested with the two-sided exact sign-flip test on per-location differences in detection counts, Holm-adjusted (Holm, 1979) over the 9 anatomies, with the paired S50 ratio and its location-bootstrap interval as the effect size.

This band, array and detector were chosen from the 24 comparisons per OPM array of each anatomy, a selection that the correction over anatomies does not control. The detection family has 48 comparisons per anatomy: two OPM arrays against Neuromag's 306 channels, its gradiometers and its magnetometers, with both detectors, in 4 depth bands. The localization family has 40: two OPM arrays; focal and patch sources at two strengths; and the dipole error, the two errors of dy-

1007 namic statistical parametric mapping (dSPM), and joint detection and localization
1008 with the study's dSPM or with the dipole.

1009 The sign-flip test uses the locations whose difference in detection counts is not
1010 zero, and p is smallest when all of them favor one array. With a band's 18 locations
1011 that smallest p is 7.6×10^{-6} , and $p < 0.05$ needs at least 6 such locations. With equal
1012 differences the test is the exact sign test. Of 18 locations, 14 must then favor one
1013 array for $p < 0.05$ ($p = 0.031$; with 13, $p = 0.096$), and 16 for the first Holm step
1014 over 9 anatomies, $p < 0.0056$ ($p = 0.0013$). Of the confirmatory 36 locations (12
1015 per orientation stratum), 25 are needed ($p = 0.029$) and 27 for the first Holm step
1016 ($p = 0.0039$).

1017 Unequal differences can move p either way: the adult's 11 locations favoring
1018 the OPM against 1 favoring Neuromag gave $p = 0.0039$, against 0.0063 had the
1019 differences been equal. Beyond 20 non-zero locations the run estimated p from
1020 20,000 random sign patterns (resolution 5×10^{-5}); the p values reported for the
1021 confirmatory run are exact (Section S7.5). These values follow from the design
1022 alone. Power, and a minimal detectable S50 ratio, depend on how detection varies
1023 between locations and were not computed. Two pilot runs on test seeds (child A,
1024 18 locations) gave 8/1 and 9/4 locations favoring the OPM/Neuromag, $p = 0.074$
1025 and 0.21, and their outputs were not kept. The confirmatory declaration describes
1026 18 locations as underpowered, a reading we do not adopt. The pilot runs prompted
1027 doubling the locations to 36 but do not estimate power.

1028 *S7.2. Results of the exploratory run*

1029 The exploratory run had one noise realization per event and an endpoint chosen
1030 after the analyses. It is superseded as evidence by the confirmatory run (main text,
1031 Section 3.6; Section S7.5) and is reported here in full. In the adult, focal spikes
1032 10–20 mm below the scalp needed 47 nAm for half detection with Neuromag's
1033 306 channels and 34 nAm with the dense array, a paired S50 ratio of 1.36 [1.10,
1034 1.58]. Every anatomy's point estimate favored the dense array in this band (S50
1035 ratios 1.18–1.65; Fig. S12A; Table S10). So did the per-location counts under a
1036 Holm correction over the 9 anatomies (largest adjusted p 0.022, child A, whose
1037 S50 interval spans 1).

1038 The anatomies are not independent replications, and the correction does not
1039 cover the choice of the endpoint from 24 comparisons per OPM array (below).
1040 The site-matched array showed no consistent advantage. The dense array's deeper
1041 spikes leaned the same way, with an established difference in only 1 of 27 deeper
1042 band-and-anatomy cells (Holm within each band; Fig. S12B). The gain lay in the
1043 weakest detected spikes.

1044 Table S10 gives the superficial band in every anatomy, with the oracle's S50
1045 ratio beside the practical detector's. The oracle is the quantity most directly tied
1046 to the detectability metric. Its ratios were larger than the practical detector's in 7

1047 of the 9 anatomies (not in the 18-month template and child C), as expected where
1048 the oracle's advantage is not diluted by the scan over times and candidates. The
1049 adult's detection curves are shown in Fig. S13.

Table S10. Detection of simulated spikes 10–20 mm below the scalp in the exploratory run. Practical detector with thresholds frozen at 1 false event per minute, 18 locations per anatomy. The endpoint was chosen after the analyses.

Head	S50 (nAm), Neuromag / dense	Ratio, dense	Locations, OPM / Neuromag	p	Holm p	Ratio, site-matched	Ratio, dense, oracle
Adult	47 / 34	1.36 [1.10, 1.58]	11 / 1	0.0039	0.015	1.16 [0.88, 1.44]	1.55 [1.22, 1.74]
School-age size	53 / 33	1.60 [1.31, 1.87]	13 / 0	0.00024	0.0017	1.27 [0.99, 1.52]	1.62
2-year size	61 / 37	1.65 [1.41, 1.87]	17 / 0	<0.0001	0.00014	1.60 [1.21, 1.82]	1.69
24-month template	43 / 30	1.44 [1.20, 1.65]	14 / 2	0.0015	0.0076	1.22 [1.00, 1.42]	1.51
18-month template	36 / 27	1.33 [1.16, 1.62]	16 / 0	<0.0001	0.00024	1.10 [1.02, 1.28]	1.30
12-month template	43 / 29	1.46 [1.21, 1.84]	14 / 1	0.00049	0.0029	1.28 [1.09, 1.46]	1.89
Child A	61 / 48	1.27 [0.93, 1.63]	11 / 4	0.022	0.022	1.05 [0.75, 1.40]	1.53
Child B	65 / 55	1.18 [1.02, 1.51]	12 / 3	0.0085	0.017	1.06 [0.87, 1.35]	1.31
Child C	52 / 36	1.43 [1.06, 1.62]	14 / 2	0.0037	0.015	1.08 [0.92, 1.31]	1.36

S50: strength for half detection (Neuromag 306 channels / dense OPM array). Ratio: paired S50 ratio Neuromag/OPM with its bootstrap interval over locations. Locations: those with more events detected by the OPM / by Neuromag. p: exact sign-flip test on per-location differences in detection counts, uncorrected and Holm-adjusted over the anatomies. Oracle: the paired S50 ratio of the dense array with the oracle detector. The adult sits at its measured head position, the other heads at top contact. OPM, optically pumped magnetometer.

1055 The individual children are the least certain. Their S50 ratios were 1.27 [0.93,
1056 1.63], 1.18 [1.02, 1.51] and 1.43 [1.06, 1.62] for A, B and C, the interval for child
1057 A including 1. Because the endpoint was chosen from 24 comparisons per OPM
1058 array, a Bonferroni correction over them (threshold 0.0021) is the stricter test. 5
1059 of the 9 anatomies passed it, the scaled adults and the templates, while the adult
1060 ($p = 0.0039$) and the three children did not. Over all 48 comparisons per anatomy
1061 (threshold 0.0010), 4 passed.

1062 The frozen thresholds gave unequal false-event rates on the held-out null data
1063 (Table S9). As a check, every detector's threshold was reset to 1 false event per
1064 minute on the held-out null data and the stored simulations were re-evaluated. The
1065 thresholds were fitted and evaluated on the same data, so this check is in-sample for
1066 the rate and does not establish equal operating points on new data; the confirmatory
1067 run uses separate evaluation null data for that. The dense array's ratios at 10–20
1068 mm stayed above 1 in every anatomy (1.21–1.63). In the adult the ratio was 1.40
1069 [1.12, 1.59] (12 locations favoring the OPM, 1 Neuromag, $p = 0.0022$), against
1070 1.36 at the frozen thresholds.

1071 The site-matched array's S50 interval lay above 1 in 4 anatomies (bounds
1072 printed as 1.00 are counted by their unrounded values). Its detection counts,
1073 however, differed (uncorrected $p < 0.05$) in only 2, the 2-year size control and the
1074 18-month template, and not in the adult ($p = 0.40$) or any child.

1075 Deeper spikes leaned the same way less consistently. The location counts fa-
1076 vored the dense array in 25 of the 27 deeper band-and-anatomy cells (1 tie, 1
1077 against), and the S50 point estimates at 20–30 mm were above 1 in 9 of 9 anatomies.
1078 Of the 6 cells with $p < 0.05$, 6 favored the OPM. None of these survived the Bonfer-
1079 roni correction over the 24 comparisons per OPM array. With the endpoint's rule,
1080 a Holm correction over the anatomies band by band, 1 survived: the adult's 45–70
1081 mm band (adjusted $p 0.035$). Which deeper cells reach $p < 0.05$ varied between
1082 the versions of the model run before the results were frozen.

1083 Of the adult's 324 superficial focal events (54 per strength), 19 were detected
1084 only by the dense array (6, 12 and 1 at 20, 40 and 160 nAm) and 1 only by Neu-
1085 romag (at 40 nAm). The dense array and Neuromag detected 7 and 1 events at 20
1086 nAm and 33 and 22 at 40 nAm. From 80 nAm up they detected nearly the same
1087 numbers (45 and 45, 50 and 49, 51 and 51). The OPM's gain was concentrated on
1088 weak superficial spikes. Its weight in a clinical recording depends on how many
1089 real discharges are that weak, which the strength grid of the design cannot tell.

1090 At the adult's 18 locations of each band, the median detectability ratio of the
1091 dense array was 1.45, 1.16, 1.09 and 1.05 from the shallowest to the deepest band.
1092 The oracle's paired S50 ratio was 1.55, 1.29, 1.09 and 1.20. The two agreed in
1093 direction in every band, and closely at 30–45 mm. The oracle ratio exceeded the
1094 detectability ratio most in the deepest band, whose interval ([1.00, 1.42]) lies just
1095 above 1 because few events are detected at the lower strengths and S50 is interpo-

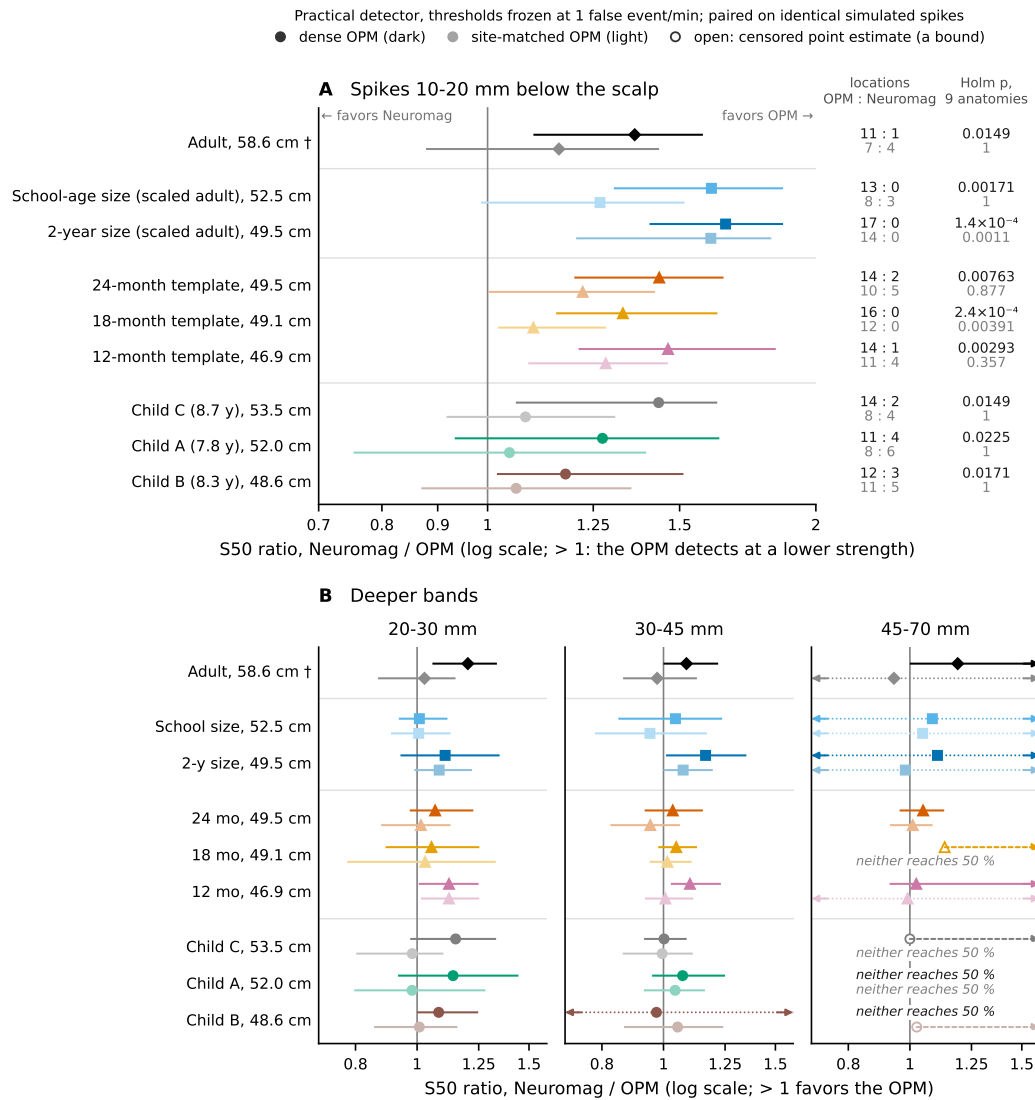


Fig. S12. Simulated interictal spikes, exploratory run. Paired S50 ratio Neuromag (306 channels) / OPM on identical simulated spikes, practical detector with thresholds frozen at 1 false event per minute; logarithmic axis with a line at 1 (above 1: the OPM detects at a lower strength). (A) Spikes 10–20 mm below the scalp, with the locations favoring each system and the location sign-flip p Holm-adjusted over the 9 anatomies for each array separately; this endpoint was chosen after the analyses. (B) The deeper bands. Dark markers: dense OPM array; light markers: site-matched OPM array. Lines: 95 % bootstrap intervals over the 18 locations per band and anatomy (324 focal events). Arrows: open interval ends (resamples outside the tested 10–320 nAm); dotted lines: intervals open at both ends; open markers: censored point estimates (an open marker with a dashed arrow: Neuromag does not reach half detection, so the ratio lies above the bound); “neither reaches 50%”: no estimate. Dagger: the adult at its measured head position; the other heads at top contact. Infant templates: O’Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016); children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025). OPM, optically pumped magnetometer.

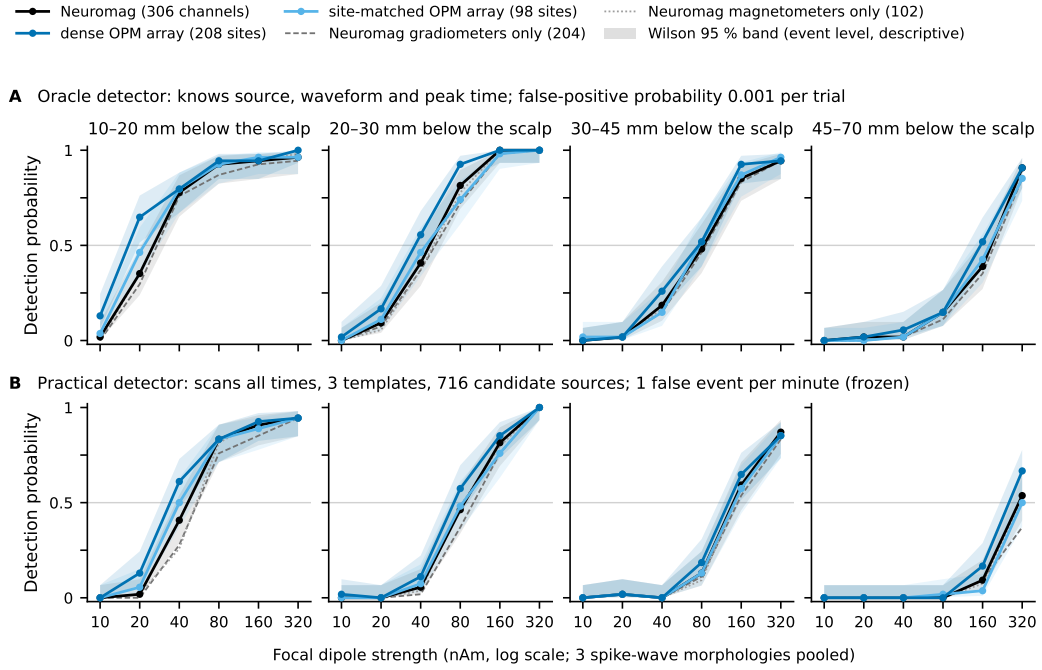


Fig. S13. Detection curves of the adult, exploratory run. Probability of detecting a focal spike against its strength (10–320 nAm, logarithmic axis; 54 events per point: 18 locations and 3 morphologies pooled) in the four depth bands, for Neuromag (306 channels), the dense OPM array (208 sites) and the site-matched OPM array (98 sites), with Neuromag’s gradiometers (dashed) and magnetometers (dotted) alone as secondary comparators. (A) The oracle detector, which knows source, waveform and peak time (false-positive probability 0.001 per trial). (B) The practical detector, which scans all times, 3 templates and 716 candidate sources, at 1 false event per minute. Its thresholds were frozen; on the held-out null data they gave 1.30, 0.95, 0.70, 1.20 and 0.65 false events per minute for Neuromag, the dense array, the site-matched array, the gradiometers and the magnetometers. Shaded: Wilson 95 % intervals at the event level, descriptive only, because the events of a point share locations; the paired comparisons treat the location as the unit. Gray horizontal line: half detection. At 10–20 mm the strength for half detection is 16, 22 and 25 nAm with the oracle and 34, 40 and 47 nAm with the practical detector (dense, site-matched, Neuromag). The adult sits at its measured head position; the noise is shared by all systems in each simulated segment. OPM, optically pumped magnetometer.

lated from them. The detectability ratio is a median of per-location ratios, whereas the S50 ratio pools the band's detection curves, so exact agreement is not expected. The oracle is the better-behaved of the two spike quantities, and the confirmatory run reports it beside the practical detector (Table S11).

S7.3. Localization

Localization was examined at 24 locations with focal and patch sources of 80 and 320 nAm, one event per source type and strength at each location (24 per condition, unstretched morphology), in independent null data. The inverse used a single-compartment boundary-element model (BEM), whereas the simulation used three layers. It also used one of 8 coregistration-error draws of 2 mm and 2° per location, the same for every array. This error displaced the adult's sources by a median of 2.9 mm, and by 2.1–3.2 mm over the anatomies. The noise covariance was a Ledoit–Wolf estimate from 5 min of null data, and the source grid had fixed orientations at 5-mm spacing (2,114–3,821 sources) and excluded the true source vertices.

Equivalent current dipoles (ECDs) were fitted with MNE-Python (Gramfort et al., 2013) at least 5 mm inside the inner skull. dSPM (Dale et al., 2000), with depth weighting 0.8 and a regularization set by an assumed signal-to-noise ratio (SNR) of 3, was computed in the study's implementation and in MNE-Python's, at the true peak sample. The error is the distance from the true source (a patch's center) to the dipole or to the dSPM peak, read on the MRI through the perturbed transform. Joint success is detection by the practical detector (thresholds from 10 min of null data) with an error of at most 10 mm. Paired error differences were tested with the Wilcoxon signed-rank test and joint success with an exact McNemar test.

Both dSPM implementations use fixed orientations on the same grid, the same noise covariance, a depth exponent of 0.8, $\lambda^2 = 1/9$, a source covariance scaled to the whitened rank and dSPM normalization. They differ in the depth weighting. The study's implementation uses the whitened gains of all of an array's channels, without a limit. MNE-Python's is limited at its default of 10 and uses the unwhitened gains of the gradiometers, or of the magnetometers where there are none (the OPM arrays). Both are reported, because the paired dSPM differences depend on this choice.

Dipole fits of detected 320-nAm focal spikes had median errors of 3.7–7.6 mm in every array and anatomy. Of 360 paired comparisons, 65 had $p < 0.05$, where 18 would be expected by chance. Most of them concerned dSPM errors (21 with the study's implementation, 27 with MNE-Python's); 6 concerned dipole errors and 11 joint detection and dSPM localization within 10 mm. Of these, 61 favored an OPM array, 3 had a zero median difference and 1 favored Neuromag. 28 concerned mostly undetected 80-nAm sources, and the two dSPM implementations

1136 share their events.

1137 In all, 8 comparisons survived a Bonferroni correction within anatomy and
1138 array, 6 of them at 80 nAm, where most sources go undetected, and 0 survived
1139 one over all 360. The adult's dSPM gain for 320-nAm patches depended on the
1140 implementation: -5.6 mm ($p = 0.054$) with the study's estimator and -1.0 mm (p
1141 $= 0.22$) with MNE-Python's.

1142 Fig. S14 shows every paired error difference with its interval. Of the 216 error
1143 comparisons, 54 had $p < 0.05$ (6 for the dipole fit, 27 for MNE-Python's dSPM, 21
1144 for the study's). 8 fell below 0.0025, the Bonferroni threshold for the 20 compar-
1145 isons of one anatomy and OPM array. At 320 nAm the dipole-fit differences lay
1146 within a few millimeters of zero in every anatomy, and the dSPM differences were
1147 negative or zero. At 80 nAm all 108 estimates rest mostly on undetected events,
1148 whose errors are those of noise, and their intervals are wide.

1149 20 medians were exactly zero, because the dSPM peaks of both systems often
1150 fall on the same vertex of the shared 5-mm grid, and 30 were positive (Neuro-
1151 mag closer). The errors include every event, detected or not, because selecting on
1152 detection would select on each system's own noise. Fig. S15 gives the share of
1153 injected events that each system both detected and localized within 10 mm. Of the
1154 72 paired comparisons per method, 0 with the dipole fit and 11 with the study's
1155 dSPM had $p < 0.05$ (exact McNemar, uncorrected), all favoring the OPM array;
1156 none survived a correction. Joint success assumes the event time is known, because
1157 localization is read at the true peak sample.

1158 Every event received a dipole. Under declared criteria, a dipole or dSPM error
1159 above 30 mm is gross, and a dipole confidence volume above 10 cm^3 is uncon-
1160 strained. Over the 864 events per system, Neuromag had 332 gross dipole errors
1161 (27 among its 456 detected events). The dense array had 310 (44 among 501 de-
1162 tected) and the site-matched array 325 (38 among 475), that is, 5.9–8.8 % of de-
1163 tected events. Gross errors of the study's dSPM among detected events numbered
1164 77 for Neuromag (17 %), 39 for the dense array (7.8 %) and 36 for the site-matched
1165 array (7.6 %).

1166 Of all gross errors (dipole and the study's dSPM, three arrays), 87 % belonged
1167 to undetected events and 73 % to 80-nAm events. The denominators differ between
1168 arrays because each array detects different events, so these rates are described, not
1169 compared. The spike study does not test the fitted helmet, so it does not show that
1170 fit causes a detection benefit in children. No localization difference was established
1171 in any anatomy.

1172 *S7.4. Head motion*

1173 Head motion and slippage of the OPM cap were bounded rather than simulated
1174 in time, on the adult and on the 24- and 12-month templates at top contact. The
1175 head was displaced in the fixed helmet by up to 10 mm or rotated about the head

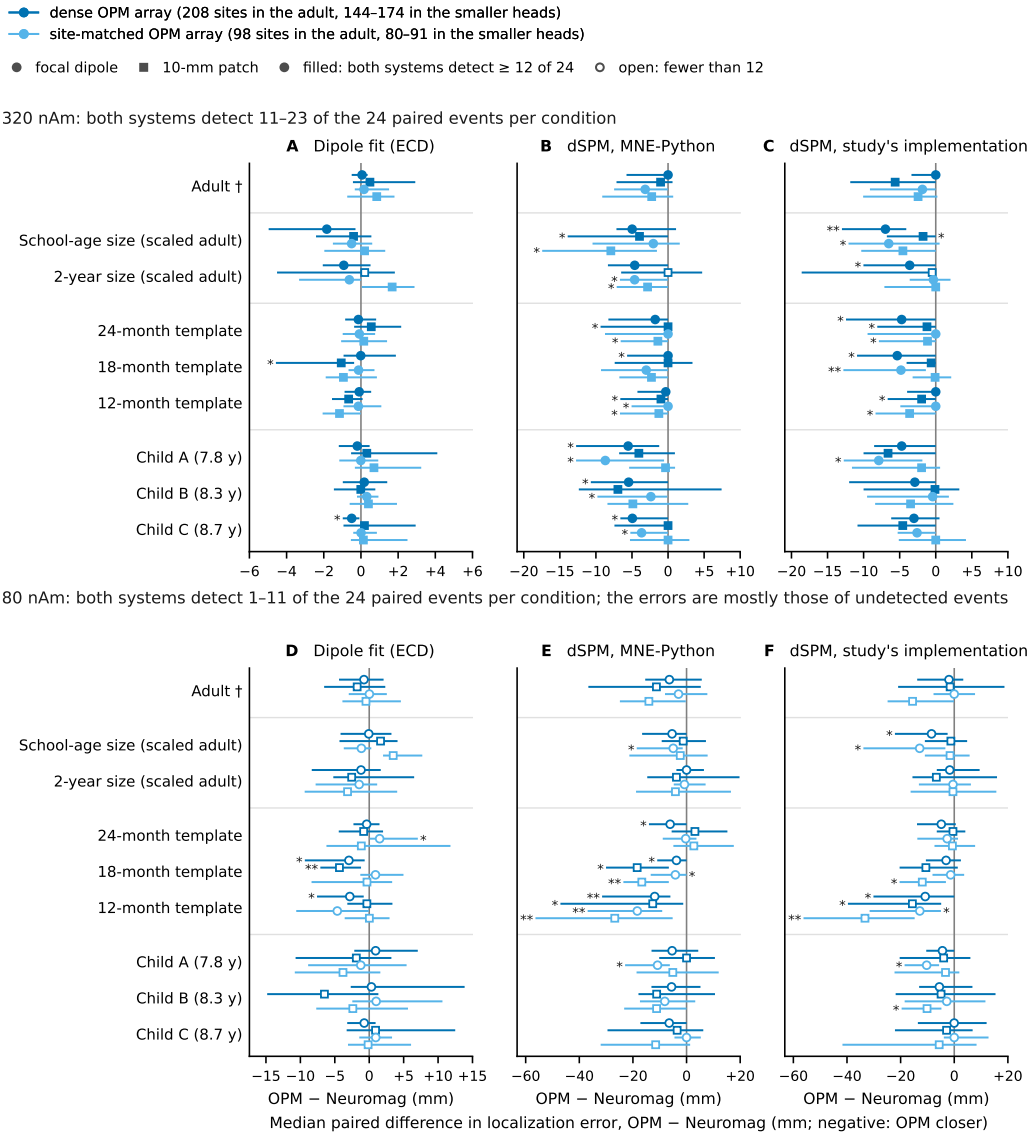


Fig. S14. Localization effect sizes, exploratory run. Median over the 24 events of a condition of the OPM array's localization error minus Neuromag's (306 channels), paired on identical simulated spikes, with 95 % bootstrap intervals over events (2,000 resamples); negative: the OPM array localizes closer to the true source. Columns: the equivalent-current-dipole (ECD) fit, dSPM through MNE-Python and dSPM through the study's implementation. (A–C) 320 nAm. (D–F) 80 nAm, where the errors are mostly those of undetected events. Dark blue: dense OPM array; light blue: site-matched OPM array; circles: focal dipoles; squares: 10-mm patches; filled markers: both systems detected at least half of the events (practical detector, thresholds frozen at 1 false event per minute); open markers: fewer. Stars beside an interval: $p < 0.05$ (one) and $p < 0.0025$ (two), Wilcoxon signed-rank, uncorrected; the second threshold is 0.05 over the 20 comparisons of one anatomy and OPM array. The x axes differ between strengths and between the dipole and dSPM columns. The errors are read at the true peak sample for every event, detected or not; dSPM peaks lie on the 5-mm grid, so paired errors often tie and medians at exactly zero are common. Dagger: the adult at its measured head position; the other heads at top contact in the adult helmet. dSPM, dynamic statistical parametric mapping; OPM, optically pumped magnetometer.

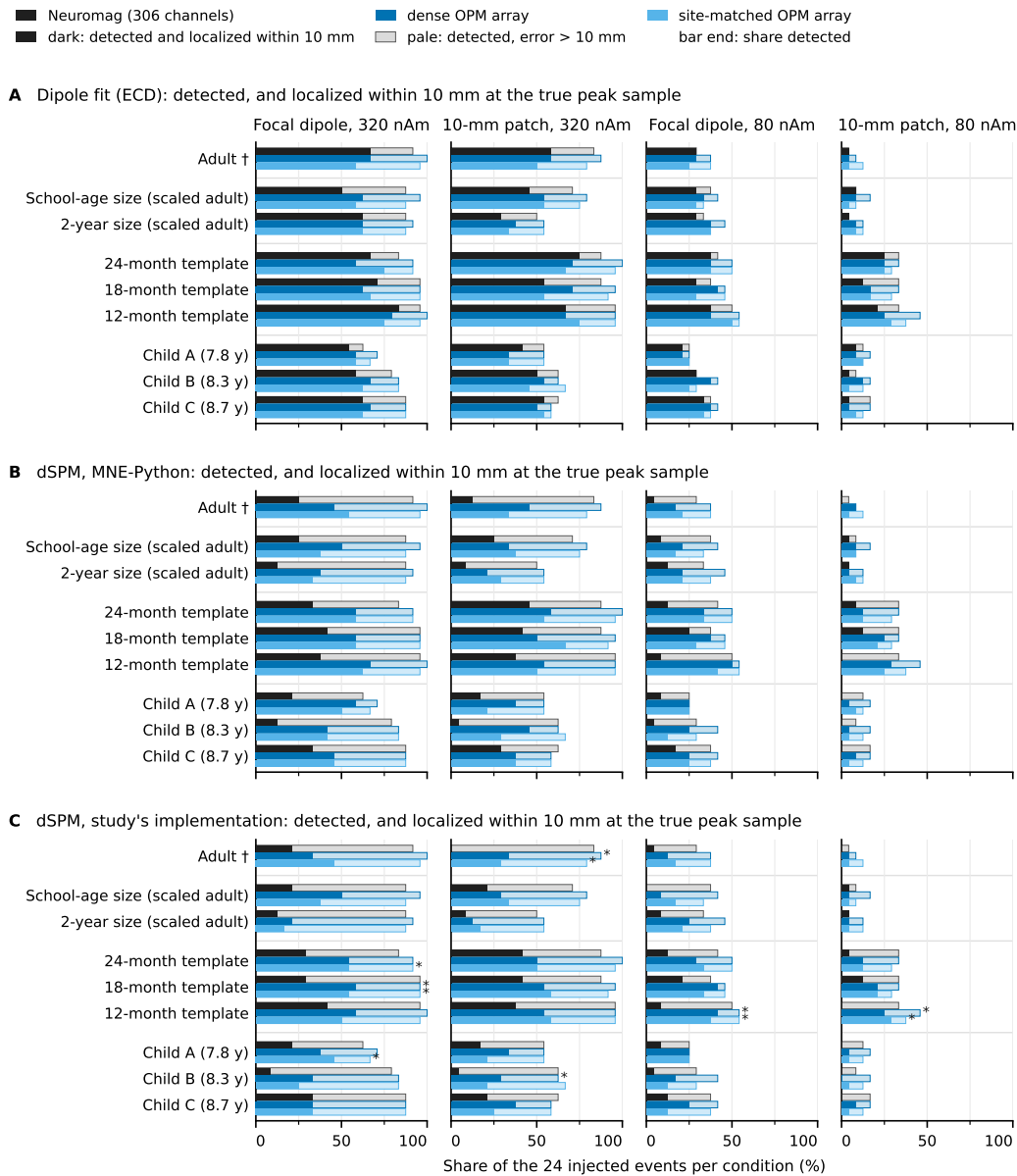


Fig. S15. Joint detection and localization, exploratory run. For each anatomy, system (Neuromag with 306 channels, the dense OPM array, the site-matched OPM array) and condition (focal dipole or 10-mm patch at 320 and 80 nAm), the bar's end is the share of the 24 injected events detected by the practical detector. Its dark part is the share both detected and localized within 10 mm, and the pale part is detected but localized farther away; both shares have the injected events as their denominator. (A) The equivalent-current-dipole (ECD) fit. (B) dSPM through MNE-Python. (C) dSPM through the study's implementation. Stars: the paired OPM minus Neuromag difference in joint success with $p < 0.05$ (exact McNemar, uncorrected; stored for the dipole fit and the study's dSPM only). Thresholds were set for 1 false event per minute on 10 min of null data and frozen (0.2–1.9 per minute on the localization study's held-out null data over the nine anatomies and three systems). Localization is read at the true peak sample, so joint success assumes the event time is known. Dagger: the adult at its measured head position; the other heads at top contact in the adult helmet. dSPM, dynamic statistical parametric mapping; OPM, optically pumped magnetometer.

1176 origin, and the dense OPM cap was rotated on the head by up to 3° , with sensors
1177 that would enter the scalp lifted. Each case was evaluated with the matched fil-
1178 ter of the reference geometry (uncompensated) and with the displaced geometry
1179 known. A sustained 10-mm drop in the fixed helmet cost Neuromag 1.64–1.72
1180 dB uncompensated and 0.36–0.39 dB with the displaced geometry known. The
1181 latter is comparable to an uncompensated 3° slip of the OPM cap (0.12–0.46 dB).
1182 A 60-s time course of exact rigid motion checked the field-artifact model only;
1183 motion in the spike simulations, and its interaction with detection thresholds, was
1184 not simulated.

1185 *S7.5. The confirmatory run*

Table S11. The confirmatory spike run per head. Focal spikes 10–20 mm below the scalp at 36 newly drawn locations per anatomy (648 focal events of 10–320 nAm; the first of 5 noise replicates).

Head	S50 (nAm), Neuromag / dense	Ratio, dense	Locations, OPM : Neuromag	Holm p	Oracle, ratio	Mismatch, ratio	Site-matched, ratio (Holm p)
Adult	50 / 35	1.45 [1.21, 1.68]	24 : 0	<0.0001	1.39 [1.22, 1.55]	1.50 [1.30, 1.67]	1.30 [1.10, 1.46] (0.035)
School-age size	45 / 31	1.44 [1.26, 1.58]	25 : 1	<0.0001	1.61 [1.46, 1.76]	1.39 [1.23, 1.52]	1.28 [1.10, 1.43] (0.048)
2-year size	54 / 37	1.47 [1.24, 1.62]	27 : 1	<0.0001	1.62 [1.40, 1.78]	1.49 [1.26, 1.67]	1.16 [1.05, 1.35] (0.0031)
24-month template	35 / 27	1.28 [1.18, 1.45]	28 : 2	<0.0001	1.51 [1.31, 1.84]	1.42 [1.26, 1.68]	1.06 [0.95, 1.17] (0.067)
18-month template ^a	41 / 29	1.43 [1.26, 1.61]	30 : 1	<0.0001	1.62 [1.41, 1.90]	1.44 [1.25, 1.65]	1.30 [1.15, 1.48] (<0.0001)
12-month template	41 / 29	1.39 [1.25, 1.56]	29 : 1	<0.0001	1.48 [1.30, 1.84]	1.43 [1.25, 1.60]	1.30 [1.15, 1.48] (0.0084)
Child A	51 / 40	1.27 [1.07, 1.38]	14 : 3	0.010	1.29 [1.09, 1.52]	1.23 [1.06, 1.41]	0.99 [0.84, 1.14] (1.0)
Child B	57 / 42	1.35 [1.15, 1.56]	21 : 2	<0.0001	1.27 [1.11, 1.52]	1.23 [1.02, 1.48]	1.08 [0.99, 1.20] (1.0)
Child C	50 / 39	1.30 [1.15, 1.43]	25 : 2	<0.0001	1.34 [1.17, 1.55]	1.34 [1.18, 1.50]	1.10 [1.00, 1.27] (0.15)

57
1186

1187 S50: strength for 50 % detection, Neuromag (306 channels) / dense OPM array, in whole nAm (the ratios use the unrounded values), practical
1188 detector with thresholds frozen for a nominal 1 false event per minute. Ratio: $S50_{\text{Neuromag}}/S50_{\text{OPM}}$, a ratio of the two location-pooled estimates with
1189 its 95 % interval from paired location resampling (1,000 resamples); above 1, the OPM detects at a lower strength. Locations: those at which the
1190 OPM/Neuromag detected more events (the others are ties). Holm p: two-sided sign-flip test on the per-location differences in detection counts, exact
1191 over all sign patterns, Holm-adjusted over the nine anatomies within each column's family (the run's own Monte Carlo estimates give the same
1192 decisions in all 25 families; Section S7.5). “<0.0001”: an exact p below the printed precision. Interval bounds printed as 1.00 are classed by their

1193 unrounded values (child C's site-matched interval, printed [1.00, 1.27], has the lower bound 0.9997 and spans 1). The dense array's
1194 practical-detector columns are the declared endpoint, the others secondary. Oracle: the oracle detector (known source, waveform and time).
1195 Mismatch: the declared mismatched detector (templates between the simulated morphologies, candidate patterns from a single-compartment model
1196 with a 2-mm / 2° coregistration error; thresholds set for the same rate). Site-matched: the site-matched OPM array with the practical detector. The
1197 adult at its measured head position, the other heads at top contact in the fixed adult helmet. Infant templates: O'Reilly et al. (2021), from the
1198 Neurodevelopmental MRI Database (Richards et al., 2016); children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025). OPM, optically
1199 pumped magnetometer. ^aThe 18-month template is classed misregistered by the MRI check (a possibly misaligned white surface, or the weak
1200 contrast of an average image; main text, Section 2.2).

1201 The confirmatory design was fixed before the run and declared in a configura-
1202 tion file of the code repository (main text, Section 2.5). Its primary result is given
1203 in the main text (Section 3.6; Fig. 8) and in Table S11. This subsection gives
1204 the declarations in full, the design as run, the Monte Carlo replicates, the realized
1205 false-event rates, the mismatch costs, the site-matched array and the results pooled
1206 over the replicates.

1207 The declaration fixed seven elements. The seeds derive from a root seed
1208 (20261005), from which one stream per anatomy and purpose is spawned. The 36
1209 locations per anatomy are newly drawn in the 10–20 mm band (12 per orientation
1210 stratum), vertex-disjoint from the exploratory ones where a stratum allows. The
1211 test is the two-sided sign-flip test, exact up to 20 non-zero location differences
1212 and otherwise estimated from 20,000 random sign patterns. The effect size is the
1213 paired S50 ratio with a location-bootstrap interval of 1,000 resamples.

1214 The null data are separate segments for the whitener, for fitting the thresholds,
1215 for the held-out check and for a 20-min evaluation of the false-event rates of the
1216 frozen and the rate-matched thresholds. Each event has 5 noise replicates, the first
1217 confirmatory and the others measuring Monte Carlo variability. The mismatched
1218 detector has templates at the geometric midpoints between the injected morpholo-
1219 gies, candidate patterns from a single-compartment BEM, and a coregistration
1220 error of 2 mm and 2° shared by all arrays.

1221 The run was made at a single code version with the configuration as declared
1222 and is complete in all nine anatomies. Every random stream was spawned from the
1223 root seed 20261005 by anatomy and purpose (10 purposes), and the noise repli-
1224 cates of an event from its event stream. The exception is the random sign patterns
1225 of the sign-flip test, which use the fixed seed 0 of the test’s implementation in ev-
1226 ery anatomy, family and replicate (Table S17). The 36 locations per anatomy (12
1227 in each of the 3 orientation strata) were drawn from pools of 109–2,249 candidate
1228 vertices per stratum. The 72 exploratory locations had been left out of the pools,
1229 so that no stratum shares a vertex with the exploratory run.

1230 The nearest exploratory location lay 2.7 mm away at the closest and a median
1231 13.6–18.6 mm away over the anatomies. The locations’ median depth was 12.9–
1232 17.4 mm; the adult’s locations lay 17 in the frontal lobe, 10 in the parietal, 8 in the
1233 temporal and 1 in the occipital lobe. Each location received 18 focal events per
1234 replicate (6 strengths of 10–320 nAm, morphologies stretched by 0.75, 1 and 1.5).
1235 This gave 648 events per replicate, 3,240 per anatomy over the five replicates and
1236 29,160 in all.

1237 The null data were 10 min for the whitener, 20 min for the thresholds (20 events
1238 above each frozen threshold), 20 min held out and 20 min of evaluation null, 70 min
1239 in all. The practical detector’s dictionary held 452–728 candidate patterns over the
1240 anatomies. The S50 intervals use 1,000 location resamples. The oracle’s realized
1241 per-trial false-positive probability on the held-out null was 0.00093–0.0011 over

1242 the anatomies and arrays, against the declared 0.001.

1243 With more than 20 non-zero location differences, the run estimated each sign-
1244 flip p from 20,000 random sign patterns (resolution 5×10^{-5}), and an estimate was
1245 zero where no sampled pattern reached the observed statistic. The differences are
1246 integer counts, so the exact p over all 2^n sign patterns follows from the distribution
1247 of the signed sum. We computed it from the stored differences for all 540 cells
1248 of anatomy and comparison; the 47 cells that the run had enumerated itself were
1249 reproduced exactly. Every p reported here and in the main text is this exact value.
1250 The run's estimates differ from it by at most 0.0073 and give the same Holm de-
1251 cision for every anatomy in all 25 families. Where an estimate was zero, the exact
1252 Holm-adjusted value reaches 0.00040.

1253 Before the run, the confirmatory endpoint code was applied to the exploratory
1254 run's stored states. It reproduced the exploratory location counts, p values and
1255 S50 ratios exactly in all 36 comparisons (nine anatomies, both OPM arrays, both
1256 detectors).

1257 The mismatch variant's templates sit at the geometric midpoints of the injected
1258 stretches (0.87 and 1.22: 2 templates), each injected morphology lying 13–18 %
1259 away from the nearest template. Its candidate patterns come from a 1-layer BEM.
1260 One coregistration error per anatomy (2 mm, 2°), shared by all arrays, displaced
1261 the candidate dictionary by a median of 1.6–4.9 mm at the spike locations over
1262 the anatomies (0.7–5.9 mm over all locations). Its thresholds were set for a nom-
1263 inal 1 false event per minute on the calibration null and frozen. Like the primary
1264 detector's, they were also re-fitted on the held-out null as a secondary analysis.

1265 In each of the five Monte Carlo replicates the endpoint passed in all nine
1266 anatomies (45 of 45 anatomy-replicate cells; Table S12). Over the replicates the
1267 ratio ranged from 1.15 to 1.49. Its SD within an anatomy was 0.041–0.089 in
1268 $\log_2$ units, that is, 2.9–6.4 % from replicate to replicate. Pooled over the replicates
1269 (3,240 events per anatomy), the ratios were 1.22–1.44 with a Holm-adjusted p of
1270 at most 0.0013. The oracle's pooled ratios were 1.35–1.60 and the mismatched
1271 detector's 1.23–1.45, each family passing in all nine anatomies. The confirmatory
1272 ratio was 0.89–1.14 times the exploratory one, below it in 6 of the 9 anatomies,
1273 and the exploratory point estimate lay within the confirmatory interval in 6.

Table S12. Replicates of the confirmatory endpoint. The declared endpoint in every noise replicate: dense OPM array against Neuro-mag’s 306 channels, practical detector, thresholds frozen for a nominal 1 false event per minute, focal spikes 10–20 mm below the scalp at 36 locations per anatomy.

Head	Ratio per replicate	SD (%)	Largest p	Confirmatory / exploratory	Pooled ratio	Pooled locations	Pooled Holm p
Adult	1.45, 1.46, 1.36, 1.39, 1.41	2.9 %	<0.0001	1.06	1.41 [1.22, 1.56]	33 / 0	<0.0001
School-age size	1.44, 1.36, 1.49, 1.49, 1.39	4.2 %	<0.0001	0.90	1.44 [1.32, 1.53]	35 / 0	<0.0001
2-year size	1.47, 1.38, 1.44, 1.43, 1.34	3.7 %	<0.0001	0.89	1.42 [1.25, 1.53]	34 / 1	<0.0001
24-month template	1.28, 1.21, 1.23, 1.21, 1.37	5.4 %	<0.0001	0.89	1.25 [1.18, 1.37]	34 / 1	<0.0001
18-month template	1.43, 1.30, 1.32, 1.49, 1.46	6.2 %	<0.0001	1.07	1.39 [1.30, 1.53]	35 / 0	<0.0001
12-month template	1.39, 1.27, 1.31, 1.32, 1.30	3.6 %	<0.0001	0.95	1.31 [1.22, 1.46]	33 / 2	<0.0001
Child A	1.27, 1.16, 1.24, 1.26, 1.19	4.0 %	0.031	0.99	1.22 [1.09, 1.33]	26 / 7	0.0013
Child B	1.35, 1.33, 1.25, 1.15, 1.29	6.4 %	0.0011	1.14	1.27 [1.12, 1.44]	26 / 6	<0.0001
Child C	1.30, 1.24, 1.27, 1.35, 1.25	3.5 %	0.00011	0.90	1.28 [1.17, 1.38]	30 / 2	<0.0001

Ratio: the paired S50 ratio Neuromag/OPM in each of the 5 noise replicates (replicate 0, the confirmatory one, first), the same events with independent noise. SD: the replicate-to-replicate spread of the ratio in percent, $2^s - 1$ with s the SD of the $\log_2$ ratio. Largest p: the largest uncorrected sign-flip p over the replicates. Confirmatory / exploratory: replicate 0’s ratio over the exploratory run’s. Pooled: all 5 replicates together (3,240 events), with its 95 % location-bootstrap interval, the locations favoring the OPM/Neuromag, and the sign-flip p Holm-adjusted over the anatomies. OPM, optically pumped magnetometer; SD, standard deviation.

1280 The frozen thresholds were set for a nominal 1 false event per minute on the
1281 calibration null (Table S13). Over the 27 anatomy-and-array combinations they re-
1282 alized 0.65–1.75 per minute on the held-out null and 0.65–2.25 on the evaluation
1283 null, the exact Poisson interval excluding the target in 5 and 5 of them. The thresh-
1284 olds matched on the held-out null realized 0.45–2.05 per minute on the evaluation
1285 null (4 excluding the target).

1286 Between the endpoint's two arrays, an approximate equal-rate test gave $p < 0.05$
1287 in 2 of the 9 anatomies with the frozen thresholds, the 24-month template and child
1288 B, once in each direction. In the 24-month template Neuromag had 2.25 against the
1289 dense array's 1.25 per minute, and in child B 0.70 against 1.40. With the matched
1290 thresholds the test gave $p < 0.05$ in 1 (school-age size). The test is a conditional
1291 binomial test on the two event counts and only indicative, because the arrays share
1292 the background and room noise. Over all 72 such tests (two OPM arrays, two
1293 detectors, two threshold rules, nine anatomies), 9 had $p < 0.05$. With the matched
1294 thresholds the endpoint's ratios were 1.27–1.50 (Holm-adjusted p at most 0.010;
1295 pooled over the replicates 1.20–1.49), against 1.27–1.47 with the frozen ones.

1296 Each system's S50 with the mismatched detector over its S50 with the primary
1297 one was 1.02–1.14 for Neuromag, 1.01–1.12 for the dense array and 0.98–1.13 for
1298 the site-matched array (Table S14). The interval lay above 1 in 5, 2 and 4 of the
1299 9 anatomies, respectively (bounds printed as 1.00 are counted by their unrounded
1300 values). The change of the S50 ratio, Neuromag's cost over the dense array's, was
1301 0.91–1.11, every interval spanning 1 (0.96–1.03 pooled over the replicates). No
1302 differential cost was established, which does not show that the two costs are equal.

1303 The mismatched detector's ratios were 1.23–1.50 with its frozen thresholds
1304 (Holm-adjusted p at most 0.026). With thresholds re-fitted on the held-out null
1305 they were 1.16–1.51. Child B then did not pass in replicate 0 (Holm-adjusted p
1306 0.072) but did pass pooled over the replicates (9 of 9 anatomies passing).

Table S13. Realized false-event rates in the confirmatory run. False events per minute, practical detector: at the frozen thresholds (set for a nominal 1 per minute on 20 min of calibration null) on the 20 min of held-out null and on the 20 min of evaluation null, and at the thresholds matched to 1 per minute on the held-out null, evaluated on the evaluation null, which no threshold saw.

Head	Neuromag (306 channels)	Dense OPM array	Site-matched OPM array	Equal-rate p, dense vs Neuromag (frozen; matched)
Adult	1.20 / 1.05 / 0.95	1.10 / 1.20 / 1.20	1.20 / 1.30 / 1.05	0.77; 0.54
School-age size	1.60 / 1.35 / 0.70	0.90 / 1.25 / 1.60	1.00 / 0.65 / 0.70	0.89; 0.011
2-year size	1.15 / 1.45 / 1.25	1.40 / 1.40 / 1.00	1.30 / 1.55 / 1.15	1.0; 0.55
24-month template	1.75 / 2.25 / 1.45	1.20 / 1.25 / 1.15	1.10 / 1.15 / 0.80	0.022; 0.49
18-month template	0.75 / 0.80 / 1.15	0.65 / 1.15 / 1.25	1.25 / 1.00 / 0.70	0.34; 0.89
12-month template	1.35 / 1.05 / 0.90	0.75 / 1.00 / 1.40	1.15 / 0.75 / 0.45	1.0; 0.18
Child A	0.95 / 0.75 / 0.80	1.15 / 0.90 / 0.90	1.40 / 1.35 / 1.05	0.73; 0.86
Child B	0.70 / 0.70 / 1.35	1.55 / 1.40 / 1.05	1.65 / 1.55 / 1.20	0.044; 0.47
Child C	1.10 / 1.40 / 1.35	1.60 / 2.15 / 1.50	0.75 / 1.65 / 2.05	0.096; 0.79

Each array's entries: held-out, frozen / evaluation, frozen / evaluation, matched. Last column: the approximate equal-rate test between the dense array and Neuromag on the evaluation null, p with the frozen and with the matched thresholds. OPM, optically pumped magnetometer.

Table S14. The declared detector-mismatch variant. Dense OPM array against Neuromag's 306 channels, replicate 0.

Head	Ratio, mismatched detector	Locations, OPM / Neuromag	Holm p	Cost, Neuromag	Cost, dense array	Ratio change	Re-fitted thresholds: ratio (Holm p)
Adult	1.50 [1.30, 1.67]	27 / 0	<0.0001	1.09 [1.01, 1.22]	1.05 [0.99, 1.17]	1.03 [0.91, 1.16]	1.48 [1.27, 1.64] (<0.0001)
School-age size	1.39 [1.23, 1.52]	28 / 2	<0.0001	1.03 [0.97, 1.11]	1.07 [0.99, 1.14]	0.97 [0.89, 1.05]	1.42 [1.26, 1.54] (<0.0001)
2-year size	1.49 [1.26, 1.67]	29 / 0	<0.0001	1.06 [1.02, 1.14]	1.05 [0.99, 1.13]	1.01 [0.93, 1.09]	1.51 [1.28, 1.69] (<0.0001)
24-month template	1.42 [1.26, 1.68]	29 / 0	<0.0001	1.14 [1.05, 1.29]	1.03 [0.98, 1.09]	1.11 [0.99, 1.27]	1.39 [1.24, 1.65] (<0.0001)
18-month template	1.44 [1.25, 1.65]	28 / 1	<0.0001	1.03 [0.98, 1.09]	1.02 [0.96, 1.10]	1.01 [0.93, 1.09]	1.38 [1.21, 1.58] (<0.0001)
12-month template	1.43 [1.25, 1.60]	28 / 2	<0.0001	1.04 [0.97, 1.10]	1.01 [0.97, 1.06]	1.02 [0.95, 1.10]	1.43 [1.26, 1.63] (<0.0001)
Child A	1.23 [1.06, 1.41]	15 / 4	0.026	1.06 [1.01, 1.12]	1.10 [1.01, 1.17]	0.97 [0.89, 1.08]	1.26 [1.10, 1.43] (0.010)
Child B	1.23 [1.02, 1.48]	20 / 7	0.0089	1.02 [0.93, 1.11]	1.12 [1.00, 1.25]	0.91 [0.79, 1.06]	1.16 [1.01, 1.39] (0.072)
Child C	1.34 [1.18, 1.50]	26 / 4	<0.0001	1.08 [1.02, 1.15]	1.05 [0.94, 1.15]	1.03 [0.94, 1.16]	1.27 [1.15, 1.44] (<0.0001)

Ratio: the paired S50 ratio Neuromag/OPM with the mismatched detector at its frozen thresholds, with its 95 % location-bootstrap interval, the locations favoring the OPM/Neuromag and the sign-flip p Holm-adjusted over the anatomies. Cost: each system's S50 with the mismatched detector over its S50 with the primary detector (above 1: the mismatch costs that system strength). Ratio change: Neuromag's cost over the dense array's, the factor by which the mismatch changes the S50 ratio. Re-fitted: the ratio with the mismatched detector's thresholds re-fitted to a nominal 1 false event per minute on the held-out null, with its Holm-adjusted p. OPM, optically pumped magnetometer.

1313 With the practical detector, the site-matched array's S50 ratios were 0.99–1.30
1314 (Table S15). The interval lay above 1 in 5 anatomies and spanned 1 in 4 (bounds
1315 printed as 1.00 are counted by their unrounded values). Holm over the anatomies
1316 passed in 5 (the adult, school-age size, 2-year size, 18-month template and 12-
1317 month template) and not in the 24-month template, child A, child B and child C.
1318 With re-fitted thresholds 5 anatomies passed, with the oracle 4 (ratios 0.99–1.43)
1319 and with the mismatched detector 6 (ratios 1.03–1.36).

1320 Pooled over the replicates, 7 anatomies passed with the practical detector (ra-
1321 tios 1.06–1.27; failing: child A and child B), 7 with the oracle and 6 with the
1322 mismatched detector. In the school-aged children the site-matched array's advan-
1323 tage was not established in replicate 0 under any detector (ratios 0.99–1.10 with
1324 the practical detector). Pooled over the replicates, child C passed with the practical
1325 detector and the oracle but not with the mismatched detector.

Table S15. The site-matched OPM array in the confirmatory run. Site-matched OPM array against Neuromag’s 306 channels (secondary family), focal spikes 10–20 mm below the scalp.

Head	Practical: ratio	Locations, OPM / Neuromag	Holm p	Re-fitted: ratio (Holm p)	Oracle: ratio (Holm p)	Mismatched: ratio (Holm p)	Pooled: ratio (Holm p)
Adult	1.30 [1.10, 1.46]	17 / 5	0.035	1.32 [1.13, 1.45] (0.017)	1.21 [1.02, 1.34] (0.23)	1.26 [1.08, 1.40] (0.0038)	1.26 [1.11, 1.36] (0.00066)
School-age size	1.28 [1.10, 1.43]	20 / 7	0.048	1.32 [1.15, 1.45] (0.0097)	1.43 [1.26, 1.57] (0.00015)	1.22 [1.09, 1.33] (0.0033)	1.27 [1.13, 1.37] (0.0015)
2-year size	1.16 [1.05, 1.35]	19 / 2	0.0031	1.12 [1.03, 1.29] (0.0097)	1.18 [1.09, 1.33] (0.00013)	1.13 [1.06, 1.25] (0.00085)	1.19 [1.10, 1.35] (<0.0001)
24-month template	1.06 [0.95, 1.17]	18 / 6	0.067	1.05 [0.94, 1.17] (0.10)	1.17 [1.02, 1.38] (0.066)	1.19 [1.04, 1.39] (0.0038)	1.11 [1.03, 1.22] (0.0054)
18-month template	1.30 [1.15, 1.48]	23 / 2	<0.0001	1.30 [1.14, 1.47] (0.00024)	1.27 [1.10, 1.44] (<0.0001)	1.36 [1.21, 1.51] (<0.0001)	1.26 [1.15, 1.40] (<0.0001)
12-month template	1.30 [1.15, 1.48]	20 / 4	0.0084	1.29 [1.14, 1.48] (0.017)	1.25 [1.04, 1.49] (0.011)	1.27 [1.12, 1.44] (0.0075)	1.21 [1.13, 1.34] (0.00062)
Child A	0.99 [0.84, 1.14]	8 / 8	1.0	0.99 [0.85, 1.14] (1.0)	1.01 [0.85, 1.20] (1.0)	1.03 [0.88, 1.18] (1.0)	1.06 [0.95, 1.19] (0.39)
Child B	1.08 [0.99, 1.20]	13 / 6	1.0	1.05 [0.95, 1.20] (1.0)	0.99 [0.87, 1.11] (1.0)	1.07 [0.97, 1.19] (1.0)	1.11 [1.01, 1.22] (0.24)
Child C	1.10 [1.00, 1.27]	15 / 6	0.15	1.12 [1.01, 1.28] (0.10)	1.12 [1.00, 1.28] (0.55)	1.06 [0.98, 1.18] (0.23)	1.12 [1.02, 1.26] (0.038)

Ratio: the paired S50 ratio Neuromag/OPM with its 95 % location-bootstrap interval. Locations: those favoring the OPM/Neuromag. Holm p: the sign-flip p Holm-adjusted over the anatomies within the column’s family; all Holm p values are exact (Section S7.5). Practical: thresholds frozen for a nominal 1 false event per minute. Re-fitted: thresholds matched on the held-out null. Oracle: known source, waveform and time. Mismatched: the declared mismatch variant at its frozen thresholds. Pooled: the practical detector over all 5 noise replicates. OPM, optically pumped magnetometer.

1331 The confirmatory run does not test the fitted helmet, the deeper bands, local-
1332 ization, the noise-model alternatives of Section S3 or a covariance mismatch of
1333 the detector. Its detector remains idealized in the ways named in the main text
1334 (Section 4.4), and the mismatch variant is one declared departure, not a model of
1335 clinical review.

1336 **S8. Parameters and analysis choices**

1337 *S8.1. Resamples, seeds and time-domain parameters*

1338 Table S16 gives the number of bootstrap resamples behind every interval fam-
1339 ily of the study, Table S17 the random seed of every simulation and resampling,
1340 and Table S18 the parameters of the time-domain model of the spike study. The
1341 confirmatory run’s resample count and seeds are included; its results are in the
1342 main text (Section 3.6) and in Section S7.

1343 The per-target result tables of the adult comparison, the smaller heads and the
1344 helmet fitted at the adult’s gap are accompanied by tables of every target’s depth
1345 below the scalp at full precision. These give each target’s membership of each
1346 depth bin, depth stratum, spike depth band and orientation stratum as half-open
1347 bins. Every count stated in the text, such as child B’s targets shallower than 10
1348 mm, can therefore be reproduced exactly; the adult’s detectabilities are given per
1349 target in the same way.

1350 *S8.2. Analysis choices made after results were known*

1351 Two noise conditions were named in the analysis configuration before any sim-
1352 ulation: sensor plus brain noise, the primary condition, and the room field pro-
1353 jected out. Whole-cortex results are reported in both conditions and regional re-
1354 sults with sensor plus brain noise. No endpoint, test or margin of the original
1355 study was declared before its analyses. The exploratory spike endpoint (the dense
1356 array against Neuromag’s 306 channels at 10–20 mm with the practical detector)
1357 was chosen from 24 comparisons per OPM array (Section S7). The regions were
1358 chosen from the epilepsy literature (Section S4), and rankings among them are
1359 exploratory. Top contact, the primary placement of the fixed helmet, was chosen
1360 after the adult’s whole-cortex results were known. The laterally centered place-
1361 ment and the helmet scaled about the laterally centered head were added after the
1362 first pediatric results were known, to remove the templates’ lateral offset; the 12
1363 source-blind placements bound the first choice (Section S5). Further analyses were
1364 added after the main results were known, without changing the model: the check
1365 of the noise model against the measured Neuromag covariance and the sensitivity
1366 analyses (Section S3), the MRI quality check of the school-aged children (Sec-
1367 tion S4) and the helmet fitted at the adult’s gap (Section S5). Each reproduces the
1368 results it builds on to within rounding. The confirmatory spike run was designed

Table S16. Bootstrap resamples behind each interval family. All intervals are percentile intervals.

Intervals	Unit resampled	Resamples
Adult: primary comparisons (all estimators), lobes, triaxial control; OPM-axis decomposition	parcel labels (70)	1,000; 1,000
Adult: sensitivity analyses (sensor noise, background, Neuromag noise, head position, scalp gap, joint grid); extended sources	parcel labels	200; 200
Adult: frequency bands	targets	200
Adult: depth, distance and orientation bins; medial wall	parcel labels present in the bin (depth bins); otherwise none (medians only)	1,000; none
Smaller heads, top contact, dense array (D, Δ , strata)	parcels of each head	1,000
Smaller heads: site-matched array; centered, back, laterally centered and true-contact placements and both helmets scaled with the head; channel-count control	parcels of each head	200; 200; 200
Smaller heads: peak-channel and mean-power SNR, shifted and rotated placements, sensitivity analyses, patches	—	none (medians only)
Spikes, exploratory runs and matched-rate re-analysis (S50, paired S50 ratio)	locations	1,000
Spikes, confirmatory run	locations	1,000
Localization: paired error differences	events	2,000
Noise-model sensitivity; helmet fitted at the adult's gap; validation against the measured covariance	parcels	1,000; 1,000 (dense array, primary helmets) or 200; 1,000
Motion thresholds	32 draws, not resampled	10th to 90th percentiles

Where a cell lists several families, the resample counts follow in the same order, separated by semicolons. D: OPM detectability relative to Neuromag (dB); Δ : a smaller head's D minus the adult's; OPM: optically pumped magnetometer; S50: the source strength for 50 % detection; SNR: signal-to-noise ratio.

Table S17. Random seeds of the simulations and resamplings.

Analysis	Seed
Adult comparison: background grid and targets; extended-source bootstrap; convergence subset; frequency bands; OPM-axis decomposition; near-mesh check; BEM-sphere check	2026; 1; 2; 2026; 2026; 11; 5
Adult comparison: depth-bin bootstrap	2026
Adapted benchmarks: Hunold et al. (2016) (sources; noise); Goldenholz et al. (2009)	2016; 7; 2009
Smaller heads: parcel bootstrap (templates and scaled adults; children); patch centers; usefulness strata; helmet fitted at the adult's gap	7; 8; 2027; 0; 2026
Noise-model sensitivity and validation against the measured covariance (parcel bootstraps and surrogates)	2026; 2026
Spikes, exploratory runs (the same in all 9 anatomies): detection; location bootstrap; localization; the localization study's held-out null data; the localization study's bootstrap of the Neuromag-subset comparisons	44; 7; 45; 46; 47
Spikes, confirmatory run: root seed, with one stream per anatomy and purpose (10 purposes) and the 5 noise replicates of every event spawned from it (key: anatomy, purpose, replicate); sign-flip Monte Carlo (fixed, every anatomy and replicate)	20261005; 0
Head motion: coupling draws; time course	2027; 2028

Where a cell lists several analyses, the seeds follow in the same order, separated by semicolons.
BEM: boundary element model; OPM: optically pumped magnetometer.

Table S18. Time-domain model of the spike study. The settings are the same in every anatomy.

Element	Setting
Cortical background	independent Gaussian series, power spectrum $1/f^x$ ($x = 1$) above 0.5 Hz and flat below; band-filtered, scaled to $s^2 a_i$
Room field	the 8 coefficients synthesized from the empty-room cross-spectra (Welch, 4,096-sample segments), scaled to the fitted covariance; not projected out
Sensor noise	white (OPM 15 fT/ $\sqrt{\text{Hz}}$); no OPM frequency response (a first-order 100-Hz response only in the band analysis)
Filter, sampling	1–40 Hz, Butterworth of order 4, zero phase, also applied to the spike templates; simulated at 600.6 Hz, every 4th sample kept (no further anti-aliasing filter): 150.2 Hz
Segments	30 s, generated 2 s longer at each end and cropped; one realization shared by all arrays; a spike every 2 s
Candidate field patterns	10-mm grid without the true sources: 716 (adult); 536, 447 (scaled adults); 517, 478, 441 (templates, oldest first); 702, 623, 671 (children A–C)
Null data	detection: whitener 10 min, thresholds 20 min (then frozen), held-out 20 min, confirmatory evaluation 20 min; localization: covariance 5 min, thresholds 10 min, held-out 10 min

$s^2 a_i$: the moment variance of background source i , the variance per unit area s^2 times the area a_i of the source's cortical cell (Section S3). OPM: optically pumped magnetometer.

1369 before it was run (main text, Section 2.5) and is the only pre-specified test of the
1370 study.

1371 The model also changed while results were known. Earlier versions took the
1372 OPM sensitive axes from the adult's stored outer-skin normals, which are nearly
1373 radial, and placed the adult's head surface outside its MRI scalp, by a median of
1374 1.1 mm; they also differed in how the OPM cell's clearance from the head was
1375 checked. The final model puts every head's surface on its MRI scalp, so that the
1376 clearance rule moves only the sites the anatomy demands and the OPM standoff
1377 is the same (6.99–7.00 mm) above every head's own MRI scalp (the templates'
1378 scalp convention differs from the adult's; Section S4). It takes the sensitive axes
1379 from that surface's own triangles (Section S3). This change to equal OPM stand-
1380 off was made after the pediatric results were known. It lowered Δ by 0.11–0.27
1381 dB in the scaled adults and templates (the school-aged children exist only in the
1382 final model). It raised the adult's projected dense ratio (1.048 with the earlier con-
1383 struction, 1.073 with the final sites and the earlier axes, 1.115 in the final model).
1384 It made the laterally centered scaled-helmet Δ of the scaled adults and templates
1385 negative. It strengthened the adult's exploratory spike result at 10–20 mm (earlier
1386 8/2 locations favoring the OPM/Neuromag, $p = 0.037$, S50 ratio 1.29; final 11/1
1387 and 1.36). Every array change redraws the noise, because an array's channel count
1388 changes the random stream. The dense array's advantage at 10–20 mm held in
1389 every version, but which deeper cells reach $p < 0.05$ did not. The site-matched
1390 array's deficit at 45–70 mm in two earlier versions (0/7 and 1/9 locations, the latter
1391 with $p = 0.016$) does not reach $p < 0.05$ in the final model (3/6 locations, $p = 0.40$).
1392 Results that depend on these choices are post hoc, the earlier results are not part of
1393 the stored results, and persistence across versions is not replication. The main text
1394 states the post hoc caveat once (Section 2.6) and reports every exploratory result
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